Supervised Learning Notes

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Chapter 1: Introduction

1. Statistical learning
   (a) Refers to a vast set of tools for understanding data. Grouped into 2 categories: Supervised vs unsupervised
   (b) Supervised learning builds a statistical model for predicting an output based on one or more inputs. Both inputs and outputs must be given to build the model. Once built the model can predict an output on any input. Target goal is clear. Many ways to quantify accuracy.
      - Want to estimate relationship
      $$Y = f(X) + \epsilon$$
      for X inputs (features), Y output (response), f an unknown function to approximate, and ϵ random error which cannot be explained.
      - Prediction is key, but statistics give certainty.
      - Statistical inference helps understand how X effects Y.
   (c) Unsupervised learning only has inputs, no outputs. The model learns relationships and structure from such data. End goal is more fuzzy. Difficult to measure success.
2. Examples: Supervised vs unsupervised learning
   (a) Supervised learning: House price from size, car MPG from weight, Twitter sentiment, bird identification, student dropout
   (b) Unsupervised learning: Types of students, running gaits, topic modeling, GoT network discovery
   (c) Which is it? Music recommender, self driving car, customer segmentation
3. Statistical learning vs Machine learning (much overlap, different POV)
   (a) Statistical learning
      - Origin: Statistics and classic regression.
      - Goal: Understand the relationship between variables and provide interpretable models by extending classical statistics to flexible, data-driven models.
      - How does X effect Y, and how confident are we in that effect?
   (b) Machine learning
      - Origin: CS and AI
      - Goal: Maximize predictive performance, often on unseen data
      - How well can we predict Y from X, regardless of how and why? Though sometimes how and why.
4. History of statistical learning
   - Early 1800s, least squares and linear regression
   - 1936, linear discriminant analysis
   - 1940s, logistic regression
   - 1970, generalized linear model
   - 1980s, nonlinear models such as trees, generative additive models, neural networks
   - 1990s, SVMs
5. Matrix basics and notation for the text

Chapter 2: Statistical learning

2.1: What is statistical learning?

1. Probability crash course: Discrete case only, concrete example of dice roll.

   (a) Sample space S is the set of all possible outcomes. Die sample space $S = \{1, 2, 3, 4, 5, 6\}$.

   (b) An event $E \subset S$ is a subset of outcomes of the sample space S. Probability of subset of equally likely outcomes is given by counts. That is, for event $E = \{1\}$ is

   $$P(E) = \frac{|E|}{|S|} = \frac{1}{6}.$$

   Probability of event $E = \{2, 4, 6\}$, an even roll is

   $$P(E) = \frac{|E|}{|S|} = \frac{3}{6} = \frac{1}{2}.$$

   (c) Basic rules:
   - $0 \leq P(E) \leq 1$
   - $P(S) = 1$
   - $P(E^c) = 1 - P(E)$
   - $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
   - For mutually exclusive events, $P(A \cup B) = P(A) + P(B)$

   (d) The conditional probability of event A given B is

   $$P(A \mid B) = \frac{P(A \cap B)}{P(B)},$$

   which can be seen from the equally likely event case and a Venn diagram.

   $$P(A \mid B) = \frac{|A \cap B|}{|B|} = \frac{|A \cap B|/|S|}{|B|/|S|} = \frac{P(A \cap B)}{P(B)}$$

   For the dice, $P(4|even) = 1/3$.

   (e) Events A and B are independent if

   $$P(A \cap B) = P(A) \cdot P(B),$$

   which can be seen from conditional probability. 2 dice or flipping 2 coins is a better example.

   (f) A discrete random variable X assigns numbers to outcomes. Apparent for dice roll outcome. Could flip 2 coins and counts heads coding outcomes as $\{0, 1, 2\}$. This allows us to enumerate outcomes bringing in use of mathematics and functions.

   (g) The probability mass function $p(x) = P(X = x)$ gives the probability that X has value x, allowing for unequally likely outcomes. Note $\sum_x p(x) = 1$. For the dice, $p(x) = \frac{1}{6}$ for all outcomes.

   (h) The expected value (mean) of discrete random variable X is

   $$\mathbb{E}[X] = \sum_x x\, p(x),$$

   which represents the long-run average outcome. For the die roll, $\mathbb{E}[X] = 3.5$.

(i) The variance of X measures spread around the mean.

where $\mu = \mathbb{E}[X]$ which can be simplified as

$$\mathrm{Var}(X) = \mathbb{E}[(X-\mu)^2] = \sum_x (x-\mu)^2\, p(x) = \cdots = E(X^2) - [E(X)]^2.$$

Variance is nice from a math perspective and good for algebraic properties. For our die, $\mathrm{Var}(X) = 2.916667$ (dice value squared).

(j) The standard deviation is the square root of variance, an alternate measure of spread.

$$\sigma = \mathrm{SD}(X) = \sqrt{\mathrm{Var}(X)}.$$

Standard deviation is messy mathematically, but it is in the same units of X. For the continuous case, this represents interpretable spread (68-95-99 rule for a normal distribution). For the die, $\sigma = 1.707825$ meaning 66% (2-5) lie within one standard deviation.

(k) Expected value of a function $g(X)$ is

$$\mathbb{E}[g(X)] = \sum_x g(x)\, p(x).$$

2. Probability crash course V2: Discrete case only, concrete example of dice roll.

(a) Sample space S is the set of all possible outcomes. Die sample space is

$$S = \{1, 2, 3, 4, 5, 6\}.$$

(b) An event $E \subset S$ is a subset of outcomes of the sample space S. For equally likely outcomes, the probability of an event is given by counts. That is, for event $E = 1$,

$$P(E) = \frac{|E|}{|S|} = \frac{1}{6}.$$

For event $E = 2, 4, 6$, an even roll,

$$P(E) = \frac{|E|}{|S|} = \frac{3}{6} = \frac{1}{2}.$$

(c) Basic rules:

- $0 \le P(E) \le 1$
- $P(S) = 1$
- $P(E^c) = 1 - P(E)$
- $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
- If A and B are mutually exclusive, then $P(A \cap B) = 0$, so

$$P(A \cup B) = P(A) + P(B).$$

(d) The conditional probability of event A given B is

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)},$$

which can be understood as restricting attention to the outcomes in B. In the equally likely case,

$$P(A \mid B) = \frac{|A \cap B|}{|B|} = \frac{|A \cap B|/|S|}{|B|/|S|} = \frac{P(A \cap B)}{P(B)}.$$

For the dice,

$$P(4 \mid \text{even}) = \frac{1}{3}.$$

(e) Events A and B are independent if

$$P(A \cap B) = P(A)P(B),$$

which can be seen from conditional probability. Two dice or flipping two coins provide simple examples of independent events.

(f) A discrete random variable X assigns numbers to outcomes. For a die roll, the outcome itself can be the random variable. For example, if two coins are flipped, X could count the number of heads, with possible values $0, 1, 2$. This allows us to use mathematics and functions to describe random phenomena.

(g) The probability mass function (PMF)

$$p(x) = P(X = x)$$

gives the probability that the random variable X has value x, allowing for unequally likely outcomes. Note

$$\sum_x p(x) = 1.$$

For the die,

$$p(x) = \frac{1}{6}, \qquad x = 1, \ldots, 6.$$

(h) The expected value (mean) of a discrete random variable X is

$$\mathbb{E}[X] = \sum_x x\, p(x),$$

which represents the long-run average outcome. For the die,

$$\mathbb{E}[X] = 3.5.$$

(i) The variance of X measures spread around the mean.

$$\text{Var}(X) = \mathbb{E}[(X - \mu)^2] = \sum_x (x - \mu)^2 p(x) = \mathbb{E}[X^2] - [\mathbb{E}[X]]^2,$$

where $\mu = \mathbb{E}[X]$. Variance measures the average squared distance from the mean. For the die,

$$\text{Var}(X) = \frac{35}{12} \approx 2.916667.$$

(j) The standard deviation is the square root of variance, an alternate measure of spread.

$$\sigma = \mathrm{SD}(X) = \sqrt{\mathrm{Var}(X)}.$$

Unlike variance, standard deviation is measured in the same units as X. For the die,

$$\sigma = \sqrt{\frac{35}{12}} \approx 1.707825.$$

The die outcomes $2, 3, 4, 5$ are within one standard deviation of the mean, so $4/6 \approx 67\%$ of outcomes fall within one standard deviation. The familiar 68-95-99.7 rule applies specifically to a normal distribution.

(k) The expected value of a function $g(X)$ is

$$\mathbb{E}[g(X)] = \sum_x g(x)\, p(x).$$

For example,

$$\mathrm{Var}(X) = \mathbb{E}[(X - \mu)^2].$$

(l) The conditional expectation of Y given $X = x$ is

$$\mathbb{E}[Y \mid X = x].$$

It represents the average value of Y among observations for which $X = x$. For example, if X is horsepower and Y is fuel efficiency, then

$$\mathbb{E}[Y \mid X = 200]$$

is the average fuel efficiency for cars with 200 horsepower.

More generally, $\mathbb{E}[Y \mid X]$ is a function of X. It gives the expected value of Y for each possible value of X and can therefore be viewed as a prediction of Y based on X.

(m) Conditional expectation provides the theoretical foundation for regression. Suppose we want to predict Y from X using squared prediction error. For a particular value $X = x$, consider predicting Y with a number a. The expected squared prediction error is

$$\mathbb{E}[(Y - a)^2 \mid X = x].$$

This quantity is minimized when

$$a = \mathbb{E}[Y \mid X = x].$$

Thus, the best possible prediction of Y from $X = x$, when using squared error, is

$$\boxed{\mathbb{E}[Y \mid X = x]}.$$

In other words, the regression function

$$f(x) = \mathbb{E}[Y \mid X = x]$$

is the theoretical function that gives the best prediction of Y from X under squared error loss.

In practice, we generally do not know the true function $f(x)$. Instead, we use observed data to estimate it. Linear regression assumes a particular form,

$$f(x) \approx \beta_0 + \beta_1 x,$$

and estimates β_0 and β_1 from the data.

With multiple predictors,

$$f(x_1, \ldots, x_p) = \mathbb{E}[Y \mid X_1 = x_1, \ldots, X_p = x_p] \approx \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p.$$

Thus, regression can be viewed as estimating the conditional mean of a response given one or more predictors.

(n) For classification, the same idea leads to conditional probabilities rather than conditional means. For a binary response $Y \in 0, 1$,

$$\mathbb{E}[Y \mid X = x] = P(Y = 1 \mid X = x).$$

Thus, the conditional expectation is also the probability that $Y = 1$ given $X = x$.

For example,

$$P(\text{Survived} = 1 \mid \text{Age} = 20)$$

is the probability that a 20-year-old passenger survived. Logistic regression models this conditional probability.

This gives a common framework for supervised learning:

$$\boxed{\text{Regression: estimate } \mathbb{E}[Y \mid X]}$$

$$\boxed{\text{Classification: estimate } P(Y = k \mid X)}$$

The goal of supervised learning is to use observed data to estimate these unknown relationships between predictors X and a response Y.]

2.1: What is statistical learning?

1. Statistical learning terminology and notation (regression case)

   (a) Quantitative output variable Y (dependent variable)

   (b) p input variables $X_1, \ldots, X_p$ (independent variable)

   (c) Writing as a single vector $X = (X_1, \ldots, X_P)$, we aim to identify relationship

   $$Y = f(X) + \epsilon$$

   where f is a fixed but unknown function and ϵ is a random error term independent of X with mean zero.

   (d) Essentially, statistical learning is a set of approaches to estimate f for the purpose of both prediction and inference.

2. Prediction:

   - For $\hat{f}$ an known Why estimate of f and any input X, we predict output Y as

   $$Y \approx \hat{Y} = \hat{f}(X).$$

   - In this way $\hat{f}$ is treated as a black box where we are only concerned with output prediction.
   - Predictions have 2 sources of error: reducible and irreducible
   - Reducible error comes from $\hat{f} \approx f$ because this can be potentially reduced.
   - Irreducible error comes from ϵ because this cannot be reduced due to the inherent $X - Y$ relationship.
   - Prediction error in terms of expectation: Treating $f(X)$ and $\hat{f}(X)$ as constant and ϵ as a random variable with mean zero,

   $$\begin{aligned}
   \mathbb{E}(Y - \hat{Y})^2 &= \mathbb{E}(f(X) + \epsilon - \hat{f}(x))^2 \\
   &= \mathbb{E}(f(X) - \hat{f}(X))^2 + 2\mathbb{E}(f(X) - \hat{f}(X))\epsilon + \mathbb{E}[\epsilon^2] \\
   &= (f(X) - \hat{f}(X))^2 + 2(f(X) - \hat{f}(X))\mathbb{E}(\epsilon) + \mathrm{Var}(\epsilon) \\
   &= (f(X) - \hat{f}(X))^2 + \mathrm{Var}(\epsilon) \\
   &= \text{Reducible error} + \text{Irreducible error.}
   \end{aligned}$$

   Main property used here is linearity of expectation.

3. Inference:

   - Inference aims to explain the relationship between Y and $X_1, \ldots, X_p$ rather than predict.

4. How to estimate f?

   (a) Use **training data** to estimate f. Training data has independent and dependent variable pairs which are known.

   $$\{(x_1, y_1), \ldots, (x_n, y_n)\} \quad \text{where} \quad x_i = (x_{i1}, \ldots, x_{ip})^T$$

   (b) **Parametric methods** assume a model formula with parameters to estimate.

   - Step 1: Assume a functional form of f. For example, linear regression would give

   $$f(X) \approx \hat{f}(X) = \beta_0 + \beta_1 X_1 + \ldots \beta_p X_p$$

   where we need to estimate constant coefficients β_i rather than the entire function f.

   - Use the training data to fit the model.

   $$Y \approx \beta_0 + \beta_1 X_1 + \ldots \beta_p X_p$$

   via a computational method such as ordinary least squares.

   - Choosing a form of f has drawbacks. Too simple is easy to interpret but may **underfit** the test data resulting in low accuracy. Too complex may have high accuracy but risks **overfitting** the test data and will not generalize well to unseen data.
   - **Non-parametric methods** do not assume a form of f. Instead rules are created which moves f closer to the test data while balancing over and underfitting. Decision trees are a simple example.

5. Supervised vs unsupervised learning

(a) Supervised learning uses independent variables (X) to estimate a dependent variable (Y). A model is built using labeled training data (supervised). Goal is to understand the $X-Y$ relationship better.

(b) Unsupervised learning discovers patterns or structure in data without any labeled outputs (unsupervised). Goal is to better understand data in general.

6. Regression vs classification (both within supervised learning)

2.2: Assessing model accuracy

1. No free lunch theorem: In statistical learning (and ML), no single model works best on all problems. Instead we need to compare model performance and pick the best for the situation.

2. Measuring quality of fit:

   (a) Mean square error (MSE), most common measure:

   $$MSE_{train} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{f}(x_i))^2$$

   With this notation, MSE_{train} is measured on train data.

   (b) Test MSE, for unseen test data (x_0, y_0):

   $$MSE_{test} = \frac{1}{n}\sum_{i=1}^{n}(y_{0i} - \hat{f}(x_{0i}))^2 = Ave(y_0 - \hat{f}(x_0))^2$$

   (c) Fit balance: "U" shape test data fit curve as model flexibility increases
   - Simple models have both high train error and higher test error (underfitting)
   - Complex models have low train error but higher test error (overfitting)
   - Middle complexity models balance the 2 (just right)
   - The degrees of freedom in $\hat{f}$ determine the model flexibility

   (d) Cross validation is a more robust approach where train data is subsetted into many train-test pairs.

3. Bias-variance tradeoff:

   (a) Expected test MSE can be decomposed into **variance** of $\hat{f}$, squared **bias** of $\hat{f}$, and variance of error ϵ. That is,

   $$\mathbb{E}\left[(Y_0 - \hat{f}(x_0))^2\right] = \underbrace{\text{Var}(\hat{f}(x_0))}_{\text{variance}} + \underbrace{\left[\text{Bias}(\hat{f}(x_0))\right]^2}_{\text{squared bias}} + \underbrace{\text{Var}(\varepsilon)}_{\text{irreducible error}}$$

   where

   $$\text{Bias} = \mathbb{E}\left[\hat{f}(x_0)\right] - f(x_0).$$

   (b) Proof: Let $Y_0 = f(x_0) + \varepsilon$, where $\mathbb{E}[\varepsilon] = 0$ and $\text{Var}(\varepsilon) = \sigma^2$. We want to compute:

   $$\mathbb{E}\left[(Y_0 - \hat{f}(x_0))^2\right] = \mathbb{E}\left[(f(x_0) + \varepsilon - \hat{f}(x_0))^2\right]$$

   $$= \mathbb{E}\left[(f(x_0) - \hat{f}(x_0) + \varepsilon)^2\right]$$

$$= \mathbb{E}\left[(f(x_0) - \hat{f}(x_0))^2 + 2(f(x_0) - \hat{f}(x_0))\varepsilon + \varepsilon^2\right]$$

$$= \mathbb{E}\left[(f(x_0) - \hat{f}(x_0))^2\right] + 2\mathbb{E}\left[(f(x_0) - \hat{f}(x_0))\varepsilon\right] + \mathbb{E}\left[\varepsilon^2\right]$$

Since ε is independent of $\hat{f}(x_0)$ and has mean zero, the cross term vanishes:

$$\mathbb{E}\left[(f(x_0) - \hat{f}(x_0))\varepsilon\right] = 0 \quad \text{and} \quad \mathbb{E}\left[\varepsilon^2\right] = \sigma^2$$

Now decompose the first term using:

$$\mathbb{E}\left[(f(x_0) - \hat{f}(x_0))^2\right] = \Big(\mathbb{E}[\hat{f}(x_0)] - f(x_0)\Big)^2 + \mathrm{Var}(\hat{f}(x_0))$$

Putting it all together:

$$\mathbb{E}\left[(Y_0 - \hat{f}(x_0))^2\right] = \underbrace{\mathrm{Var}(\hat{f}(x_0))}_{\text{variance}} + \underbrace{\Big(\mathrm{Bias}(\hat{f}(x_0))\Big)^2}_{\text{squared bias}} + \underbrace{\sigma^2}_{\text{irreducible error}}$$

(c) To minimize expected test error, we need a statistical learning method that has both low variance and low bias, but can never be below irreducible error.

(d) Variance refers to how much $\hat{f}$ would change with a different set of training data. Inflexible models have low variance while flexible models have high variance.

(e) Bias refers to error introduced by approximating a real-life problem by a much simpler model. Inflexible models have high bias and flexible models have low bias.

(f) Nice picture in Ng cheatsheet. Sketch u-shape test error, bias variance tradeoff.

(g) Create a table to show tradeoff.

	Variance	Bias
Low flexibility	Low	High
High flexibility	High	Low

4. The classification setting: Working example of pass/fail class as function of study hours and attendance.

(a) y_i is a category now rather than being a quantity, though the bias-variance trade-off still occurs.

(b) We seek to estimate f from training observations $\{(x_1, y_1), \ldots, (x_n, y_n)\}$.

(c) Quantify the accuracy of our estimate $\hat{f}$ via the training error rate:

$$\frac{1}{n}\sum_{i=1}^{n} I(y_i \neq \hat{y}_i)$$

where I denotes the indicator function with value 1 if $y_i \neq \hat{y}_i$ and 0 if $y_i = \hat{y}_i$. This is the same as percent correct.

(d) Likewise, test error rate for test observations of the form (x_0, y_0) is:

$$\mathrm{Ave}(I(y_0 \neq \hat{y}_0)$$

(e) The Bayes classifier:

- It is possible to show that test error rate is minimized, on average, by assigning each observation to the most likely class given its predictor values. That is, assign x_0 to class j if
$$Pr(Y = j|X = x_0)$$
is the largest.
- The Bayes decision boundary divides the sample space into one or more classes based on this conditional probability.
- The Bayes classifier produces the lowers possible test error rate given by
$$1 - E(\max_j Pr(Y = j|X))$$
where the expectation averages the probability over all values of X. Being that this rate is less than 1, this is analogous to irreducible error.
- The issue with Bayes classifier is that these conditional probabilities cannot be computed in practice. Instead, this is an unattainable gold standard we compare other methods to.

(f) K-nearest neighbors:

- Idea is to estimate the conditional probability of a Bayes classifier as an average of the K nearest training data.
$$Pr(Y = y_i|X = X_0) = \frac{1}{K} \sum_{i \in N_0} I(y_i = j)$$
Then classify according to the largest of these probabilities over all classes j.
- The choice of K is essential. Small K is overly flexible resulting in low bias but high variance. Large K is less flexible giving high bias and low variance. Later we will have guidance to the best choice of K.

Lab: Intro to R

Homework

1. For each of parts (a) through (d), indicate whether we would generally expect the performance of a flexible statistical learning method to be better or worse than an inflexible method. Justify your answer.

   (a) The sample size n is extremely large, and the number of predictors p is small.
   Answer: Flexible best - opposite of b.

   (b) The number of predictors p is extremely large, and the number of observations n is small.
   Answer: Inflexible best - high chance of some predictors being randomly associated. Avoid overfitting.

   (c) The relationship between the predictors and response is highly non-linear.
   Answer: Flexible best - inflexible leads to high bias and underfitting.

   (d) The variance of the error terms, i.e. $\sigma^2 = Var(\epsilon)$, is extremely high.
   Inflexible best - opposite of c.

2. Explain whether each scenario is a classification or regression problem, and indicate whether we are most interested in inference or prediction. Finally, provide n and p.

(a) We collect a set of data on the top 500 firms in the US. For each firm we record profit, number of employees, industry and the CEO salary. We are interested in understanding which factors affect CEO salary.

Answer: $n = 500$, $p = 3$, regression, inference.

(b) We are considering launching a new product and wish to know whether it will be a success or a failure. We collect data on 20 similar products that were previously launched. For each product we have recorded whether it was a success or failure, price charged for the product, marketing budget, competition price, and ten other variables.

Answer: $n = 20$, $p = 13$, classification, prediction.

(c) We are interested in predicting the % change in the USD/Euro exchange rate in relation to the weekly changes in the world stock markets. Hence we collect weekly data for all of 2012. For each week we record the % change in the USD/Euro, the % change in the US market, the % change in the British market, and the % change in the German market.

Answer: $n = 52$, $p = 3$, regression, prediction.

3. We now revisit the bias-variance decomposition.

(a) Provide a sketch of typical (squared) bias, variance, training error, test error, and Bayes (or irreducible) error curves, on a single plot, as we go from less flexible statistical learning methods towards more flexible approaches. The x-axis should represent the amount of flexibility in the method, and the y-axis should represent the values for each curve. There should be five curves. Make sure to label each one.

(b) Explain why each of the five curves has the shape displayed in part (a).

Answer:

- (squared) bias: Decreases with increasing flexibility (Generally, more flexible methods result in less bias as they better fit the training data).
- variance: Increases with increasing flexibility (In general, more flexible statistical methods have higher variance as they are more sensitive to changes in training data).
- training error: Decreases with model flexibility (More complex models will better fit the training data).
- test error: Decreases initially, then increases due to overfitting (illustration of the bias-variance tradeoff).
- Bayes (irreducible) error: fixed (does not change with model).

4. You will now think of some real-life applications for statistical learning.

(a) Describe three real-life applications in which classification might be useful. Describe the response, as well as the predictors. Is the goal of each application inference or prediction? Explain your answer.

Solution:

- Coffee machine cleaned? (clean (Y/N) predicted by day of week, person assigned), inference.
- Beer type (IPA, pilsner etc. predicted by drink color, taste, etc), prediction.

(b) Describe three real-life applications in which regression might be useful. Describe the response, as well as the predictors. Is the goal of each application inference or prediction? Explain your answer.

Solution:

- Amount of bonus paid (profitability, client feedback), prediction.

- Person's height, prediction.
- House price, inference.

(c) Describe three real-life applications in which cluster analysis might be useful.
Solution:
- RNAseq tumour gene expression data.
- SNPs in human populations.
- Frequencies of mutations (with base pair context) in somatic mutation data.

5. What are the advantages and disadvantages of a very flexible (versus a less flexible) approach for regression or classification? Under what circumstances might a more flexible approach be preferred to a less flexible approach? When might a less flexible approach be preferred?

   Solution: Inflexible is more interpretable, fewer observations required, can be biased if the underlying function is over complex. Flexible can overfit (high error variance). In cases where we have high n or non-linear patterns, flexible will be preferred.

6. Describe the differences between a parametric and a non-parametric statistical learning approach. What are the advantages of a parametric approach to regression or classification (as opposed to a non-parametric approach)? What are its disadvantages?

   Solution: Parametric uses (model) parameters. Parametric models can be more interpretable as there is a model behind how data is generated. However, the disadvantage is that the model might not reflect reality. If the model is too far from the truth, estimates will be poor and more flexible models can fit many different forms and require more parameters (leading to overfitting). Non-parametric approaches do not estimate a small number of parameters, so a large number of observations may be needed to obtain accurate estimates.

7. The table below provides a training data set containing six observations, three predictors, and one qualitative response variable.

Obs.	X_1	X_2	X_3	Y
1	0	3	0	Red
2	2	0	0	Red
3	0	1	3	Red
4	0	1	2	Green
5	-1	0	1	Green
6	1	1	1	Red

   Suppose we wish to use this data set to make a prediction for Y when $X_1 = X_2 = X_3 = 0$ using K-nearest neighbors.

   (a) Compute the Euclidean distance between each observation and the test point, $X_1 = X_2 = X_3 = 0$.

   (b) What is our prediction with $K = 1$? Why?
   Solution: Green (based on data point 5 only)

   (c) What is our prediction with $K = 3$? Why?
   Solution: Red (based on data points 2, 5, 6)

   (d) If the Bayes decision boundary in this problem is highly non-linear, then would we expect the best value for K to be large or small? Why?
   Solution: Small (high k leads to linear boundaries due to averaging)

Chapter 3: Linear regression

3.1: Simple linear regression

1. We predict a quantitative response Y from a single predictor variable X assuming a linear relationship
$$Y \approx \beta_0 + \beta_1 X$$
where the intercept and slope (β_0, β_1) are learned coefficients or parameters.

2. We use our training data to produce estimates $\hat{\beta}_0, \hat{\beta}_1$ giving the prediction
$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x.$$

3. Estimating the coefficients:
   (a) Given the training set of n observation pairs
   $$(x_1, y_1), \ldots, (x_n, y_n),$$
   our goal is to obtain coefficient estimates $\hat{\beta}_0, \hat{\beta}_1$ such that the resulting line is as close to the training data as possible.
   (b) There are many ways to measure closeness, but the most natural and common choice is to minimize least squares.
   (c) Denoting the ith residual as
   $$e_i = y_i - \hat{y}_i$$
   where
   $$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i,$$
   define the residual sum of squares as
   $$RSS = e_1^2 + \cdots + e_n^2.$$
   (d) Using minimization via calculus, we can compute explicitly the least squares coefficients as follows
   $$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \overline{x})(y_i - \overline{y})}{\sum_{i=1}^n (x_i - \overline{x})^2} = \frac{Cov(X,Y)}{Var(X)}$$
   $$\hat{\beta}_0 = \overline{y} - \hat{\beta}_1 \overline{x}$$
   (e) Calculus proof: (also a linear algebra approach)

Model: $y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \quad i = 1, \ldots, n,$

RSS: $\mathrm{RSS}(\beta_0, \beta_1) = \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2.$

$$\frac{\partial \mathrm{RSS}}{\partial \beta_0} = -2\sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i) = 0 \quad \Rightarrow \beta_0 = \bar{y} - \beta_1 \bar{x},$$

$$\frac{\partial \mathrm{RSS}}{\partial \beta_1} = -2\sum_{i=1}^n x_i (y_i - \beta_0 - \beta_1 x_i) = 0 \quad \Rightarrow \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) = \beta_1 \sum_{i=1}^n (x_i - \bar{x})^2.$$

$$\Rightarrow \quad \hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2},$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}.$$

4. Assessing the accuracy of the coefficient estimates:

   (a) Recall we assume a $X - Y$ relationship of the form $Y = f(X) + \epsilon$ where ϵ is a mean-zero random error term. Then, for linear regression this becomes

   $$Y = \beta_0 + \beta_1 X + \epsilon$$

   where we think of β_0 as the expected value of Y when $X = 0$, and β_1 is the average increase in Y per unit increase in X. We aim to quantify the certainty around these expected values.

   (b) Basic example to connect to the notion of bias: Single mean population mean μ approximated by averaging many sample means $\overline{x}_i$, call is $\hat{\mu}$. Assume all samples of the same size drawn in the same way (iid).

   - We say that $\hat{\mu}$ is an unbiased approximation of μ because different samples can either cause $\hat{\mu}$ to be larger or smaller than μ. For a very large sample, it will be nearly the same as the population mean μ. This is related to the CLT. Draw the normal distribution for big small sample sizes.
   - Recall that we define bias as

   $$Bias(\hat{\mu}) = E[\hat{\mu}] - \mu = \mu - \mu = 0$$

   meaning that we have an unbiased estimate here because the CLT ensures for large number of samples $E[\hat{\mu}] = \mu$ giving zero bias. Can show this using iid assumption.
   - The standard error of $\hat{\mu}$ has a well known formula:

   $$Var(\hat{\mu}) = SE(\hat{\mu})^2 = \frac{\sigma^2}{n}$$

   where σ is the standard deviation of each of the samples X. Derivation below.

   $$E[\hat{\mu}] = E\left[\frac{1}{n}\sum_i x_i\right] = \frac{1}{n}\sum_i E[y_i] = \frac{1}{n}\sum_i \mu = \mu$$

   using linearity of expectation and

   $$Var(\hat{\mu}) = Var\left(\frac{1}{n}\sum_i \overline{x}_i\right) = \frac{1}{n^2} Var\left(\sum_i \overline{x}_i\right) = \frac{1}{n^2}\sum_i Var(\overline{x}_i) = \frac{n\sigma}{n^2} = \frac{\sigma}{n}$$

   using square property of variance.
   - The SE formula tells us the average amount $\hat{\mu}$ differs from μ as well as the effect of the sample size n. As n increases, the SE decreases.

   (c) Back to linear regression: SE formulas for the coefficient estimates.

   - Formulas can be derived.

   $$SE(\hat{\beta}_1)^2 = \frac{\sigma^2}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

   $$SE(\hat{\beta}_0)^2 = \sigma^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{\sum_{i=1}^n (x_i - \bar{x})^2}\right)$$

   where $\sigma^2 = Var(\epsilon)$ is the variance of errors, or estimated by $\hat{\sigma}^2 = \frac{1}{n-2}\sum_{i=1}^n (y_i - \hat{y}_i)^2 = \frac{RSS}{n-2}$.

- Note that $SE(\hat{\beta}_1)$ is smaller when the x_i's are more spread out. Intuitively we have more leverage to estimate the slope in this case.
- Also, $SE(\hat{\beta}_0)$ would be the same as $SE(\hat{\mu})$ if $\overline{x} = 0$ in which case β_0 would equal $\overline{y}$.
- Generally σ is unknown and estimated by the residual standard error (RSE) given above in terms of the residual sum of squares (RSS).
- Standard errors give 95% confidence intervals.

$$\hat{\beta}_1 \pm 2 \cdot SE(\hat{\beta}_1), \quad \hat{\beta}_0 \pm 2 \cdot SE(\hat{\beta}_0)$$

- Standard errors can be used to conduct hypothesis tests.

$$H_0: \text{ There is no relationship between } X \text{ and } Y$$

$$H_a: \text{ There is a relationship between } X \text{ and } Y$$

which translates to

$$H_0 : \beta_1 = 0$$
$$H_a : \beta_1 \neq 0.$$

Note, the null hypothesis translates to $Y = \beta_0 + \epsilon$ which is unrelated to X. We test with a t-statistic as with a single mean via a t-distribution with $n-2$ degrees of freedom.

$$t = \frac{\hat{\beta}_1 - 0}{SE(\hat{\beta}_1}$$

5. Assessing the accuracy of the model:
   (a) Here we quantify the extent to which the model fits the data.
   (b) Residual standard error (RSE):
   - RSE is an estimate of the standard deviation of error term ϵ. That is, the average amount the response will deviate from the true regression line.
   - RSE is computed as

$$RSE = \sqrt{\frac{1}{n-2} RSS} = \sqrt{\frac{1}{n-2} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2}$$

   where RSS denotes the residual sum of squares.
   - The units for RSE are the same as with Y making this measure interpretative.
   (c) R^2 statistic:
   - This measure normalizes to remove units to make fair model comparisons. It takes the form of a proportion of variance explained.

$$R^2 = \frac{TSS - RSS}{TSS} = 1 - \frac{RSS}{TSS}$$

   where $TSS = \sum_{i=1}^{n} (y_i - \overline{y})^2$ denotes the total sum of squares.
   - TSS is the total variance of Y and can be thought of as the amount of variability prior to preforming regression.
   - RSS is the total variance of the regression model.
   - $TSS - RSS$ is the remaining variance after regression.
   - For the simple linear regression case we have that $R^2 = r^2$ where

$$r = \text{Cor}(X, Y) = \frac{\sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n} (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^{n} (y_i - \bar{y})^2}}$$

   though this does not extend to simple linear regression.

3.2: Multiple linear regression

1. **The multiple linear regression model.** Here we consider many predictors X_j of a response variable Y.

   (a) Instead of fitting many simple linear regression models, combine the predictors into one model:

   $$Y = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p + \epsilon.$$

   Equivalently,

   $$E(Y \mid X_1, \ldots, X_p) = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p.$$

   (b) β_j represents the expected change in Y associated with a one-unit increase in X_j, **holding the other predictors fixed**.

   (c) Given estimates $\hat{\beta}_j$, predictions are

   $$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_{i1} + \cdots + \hat{\beta}_p x_{ip}.$$

   (d) The least-squares estimates minimize

   $$RSS = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2.$$

   (e) Explicit formulas for the individual $\hat{\beta}_j$ are cumbersome, but in matrix form:

   $$\hat{\boldsymbol{\beta}} = (X^T X)^{-1} X^T y.$$

2. **Question 1: Is at least one predictor useful for predicting Y?**

   (a) Test

   $$H_0 : \beta_1 = \cdots = \beta_p = 0$$

   versus

   $$H_a : \text{at least one } \beta_j \neq 0.$$

   (b) Use the F-statistic

   $$F = \frac{(TSS - RSS)/p}{RSS/(n-p-1)},$$

   where

   $$TSS = \sum_{i=1}^{n} (y_i - \bar{y})^2, \qquad RSS = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2.$$

(c) Since

$$TSS = SSR + RSS,$$

where

$$SSR = \sum(\hat{y}_i - \bar{y})^2,$$

the F-statistic can also be written

$$F = \frac{SSR/p}{RSS/(n-p-1)}.$$

It compares explained variation per predictor to unexplained variation per residual degree of freedom.

(d) Under H_0,

$$F \sim F_{p,n-p-1}.$$

Large values of F provide evidence against H_0.

(e) t-statistics and p-values can also be calculated for individual coefficients:

$$t_j = \frac{\hat{\beta}_j}{SE(\hat{\beta}_j)}.$$

3. **Question 2: Do all predictors help predict Y, or only some?**

(a) Finding a useful subset of predictors is called **variable selection**.

(b) There are 2^p possible subsets, so exhaustive search becomes impractical as p increases.

(c) Common approaches:

- **Forward selection:** Start with no predictors and add predictors one at a time according to a specified criterion.
- **Backward selection:** Start with all predictors and remove predictors one at a time according to a specified criterion.
- **Stepwise selection:** Iteratively add and remove predictors.

(d) Selection criteria include Mallows' C_p, AIC, BIC, adjusted R^2, and cross-validation error.

(e) Variable selection can produce different models for different samples and can lead to overly optimistic estimates of model performance.

4. **Question 3: How well does the model fit the data?**

(a) Two common measures are residual standard error (RSE) and R^2.

(b) Residual standard error:

$$RSE = \sqrt{\frac{RSS}{n-p-1}}.$$

It estimates the standard deviation σ of the error term.

(c) Adding predictors cannot increase RSS, but RSE can increase because the residual degrees of freedom $n-p-1$ decreases.

(d) The coefficient of determination is

$$R^2 = 1 - \frac{RSS}{TSS} = \frac{SSR}{TSS}.$$

It represents the proportion of variation in Y explained by the fitted model.

(e) With an intercept,

$$R^2 = \text{Cor}(Y, \hat{Y})^2.$$

(f) Adding predictors can never decrease R^2. Adjusted R^2 accounts for the number of predictors.

5. **Question 4: Given predictor values, what response should we predict and how accurate is the prediction?**

(a) We estimate the conditional mean

$$f(X) = E(Y \mid X).$$

Because $f(X)$ is unknown, $\hat{f}(X)$ has estimation uncertainty.

(b) A **confidence interval** quantifies uncertainty about the mean response:

$$E(Y \mid X = x).$$

(c) A **prediction interval** quantifies uncertainty about a new observation:

$$Y_{\text{new}} \mid X = x.$$

(d) Prediction intervals are wider because they include both estimation uncertainty and irreducible error ϵ.

(e) The assumed linear model may also be incorrect, producing model bias that is not captured by these intervals.

3.3: Other considerations in the regression model

1. **Qualitative (categorical) predictors.**

(a) Categorical variables require special encoding.

(b) For two levels, use a dummy variable:

$$X = \begin{cases} 0 & \text{reference category} \\ 1 & \text{other category.} \end{cases}$$

For

$$Y = \beta_0 + \beta_1 X + \epsilon,$$

β_0 is the mean response for the reference category and β_1 is the difference between the two categories.

(c) For k categories, use $k-1$ dummy variables. The omitted category is the reference/baseline category.

2. **Extensions of linear models.**

   (a) **Interactions:** An additive model assumes the contribution of each predictor is separate:

   $$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \epsilon.$$

   An interaction allows the association between one predictor and Y to depend on another:

   $$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_1 X_2 + \epsilon.$$

   In this model,

   $$\frac{\partial E(Y \mid X_1, X_2)}{\partial X_1} = \beta_1 + \beta_3 X_2.$$

   (b) **Polynomial regression:** A nonlinear relationship can be modeled using powers of a predictor:

   $$Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \epsilon.$$

   Although the relationship between Y and X is nonlinear, the model is still linear in the unknown coefficients β_j.

3. **Potential problems with linear regression.**

   (a) **Nonlinearity:** The relationship between predictors and response may not be linear. Residual plots can reveal systematic patterns. Transformations or a different model may help.

   (b) **Nonconstant variance:**

   $$Var(\epsilon \mid X) \neq \sigma^2.$$

   Transformations or weighted least squares may help.

   (c) **Correlated errors:**

   $$Cov(\epsilon_i, \epsilon_j) \neq 0.$$

   This is especially important with time series, spatial data, longitudinal data, or clustered observations.

   (d) **Outliers:** Observations with unusually large residuals can strongly affect the fitted model. Removing observations should require justification beyond improving model fit.

   (e) **High leverage:** Observations with unusual predictor values can have substantial influence on the fitted regression coefficients. Leverage statistics help identify such observations.

   (f) **Influential observations:** Some observations can substantially change the fitted model when removed. Influence measures help identify these observations.

   (g) **Multicollinearity:** Highly correlated predictors contain overlapping information. This does not necessarily reduce predictive accuracy, but it can substantially increase the variance of individual coefficient estimates and make interpretation difficult.

   Variance inflation factors (VIFs) can help identify multicollinearity.

(h) **Association versus causation:** Regression coefficients describe conditional associations. In observational data, they generally should not be interpreted as causal effects without additional assumptions or an appropriate research design.

4. **Multicollinearity example:** predictors are strongly correlated and contain overlapping information.

   Example:

$$Price = \beta_0 + \beta_1 SqFt + \beta_2 Bedrooms + \epsilon.$$

   If $SqFt$ and $Bedrooms$ are highly correlated, there are relatively few houses that differ substantially in bedrooms while having the same square footage. Thus, estimating the effect of $SqFt$ **holding** $Bedrooms$ **fixed** is difficult.

   If

$$X_2 \approx cX_1,$$

   then the columns of X are nearly linearly dependent, so

$$X^T X$$

   is nearly singular and

$$\hat{\boldsymbol{\beta}} = (X^T X)^{-1} X^T \mathbf{y}$$

   becomes unstable.

   Consequences:

$$SE(\hat{\beta}_j) \text{ can be large,} \qquad \hat{\beta}_j \text{ can be unstable,} \qquad t\text{-tests can be nonsignificant.}$$

   The overall model can still predict well; the main problem is determining the **individual effects** of correlated predictors.

   A common diagnostic is the variance inflation factor:

$$VIF_j = \frac{1}{1 - R_j^2},$$

   where R_j^2 comes from regressing X_j on the other predictors. Large R_j^2 implies large VIF_j and greater uncertainty in $\hat{\beta}_j$.

3.4: The marketing plan

Case study summary. Maybe good for class demo. Use Carseats demo in class instead.

3.5: Comparison of linear regression with K-nearest neighbors

1. Linear regression is a parametric approach. Parameters $\beta_0, \ldots, \beta_p$ need to be found. There are many advantages here, but the main disadvantage is it makes a strong assumption on the form of $f(X)$.
2. K-nearest neighbors is non parametric. There is no assumed form of $f(X)$ giving much flexibility, but the drawback is interpretability and lack of statistical testing.
3. KNN for regression simply estimates $f(x_0)$ for new test data x_0 by averaging the K training observations which are closest. That is,

$$\hat{f}(x_0) = \frac{1}{K} \sum_{x_i \in N_0} y_i$$

4. Comparison:

Aspect	**Linear Regression**	**KNN Regression**
Model type	Parametric; assumes a linear relationship between predictors and response	Non-parametric; no assumption on functional form
Interpretability	High; coefficients have clear meaning	Low; predictions are averages of nearby points
Flexibility	Low; limited to (transformed) linear relationships	High; can capture complex non-linear patterns
Bias–Variance	Lower variance but potentially high bias if model form is wrong	Lower bias for small k but higher variance
Scalability with p	Handles moderate to high p if n is sufficient	Suffers from curse of dimensionality; needs very large n for high p
Assumptions	Linearity, independence, constant variance, normality (for inference)	Only assumes nearby points have similar responses; sensitive to scaling
When to use	Relationship is roughly linear; interpretability important	Relationship is highly non-linear; prediction accuracy prioritized

Table 1: Comparison of Linear Regression and KNN Regression

Model	**Bias**	**Variance**
Linear Regression	High if the true relationship is non-linear	Low if p is small and multicollinearity is low
KNN Regression (small k)	Low	High
KNN Regression (large k)	Higher	Lower

Table 2: Bias–variance tradeoff for Linear Regression and KNN Regression

Lab: Linear regression

Homework

1. Describe the null hypotheses to which the p-values given in Table 3.4 correspond. Explain what conclusions you can draw based on these p-values. Your explanation should be phrased in terms of 'sales', 'TV', 'radio', and 'newspaper', rather than in terms of the coefficients of the linear model.

   Solution: For intercept, that $\beta_0 = 0$ For the others, that $\beta_n = 0$ (for $n = 1, 2, 3$)

   We can conclude that that without any spending, there are still some sales (the intercept is not 0). Furthermore, we can conclude that money spent on TV and radio are significantly associated with increased sales, but the same cannot be said of newspaper spending.

2. Carefully explain the differences between the KNN classifier and KNN regression methods.

   Solution: The KNN classifier is categorical and assigns a value based on the most frequent observed category among K nearest neighbors, whereas KNN regression assigns a continuous variable, the average of the response variables for the K nearest neighbors.

3. Suppose we have a data set with five predictors, X_1 = GPA, X_2 = IQ, X_3 = Level (1 for College and 0 for High School), X_4 = Interaction between GPA and IQ, and X_5 = Interaction between GPA and Level. The response is starting salary after graduation (in thousands of dollars). Suppose we use least squares to fit the model, and get $\hat{\beta}_0 = 50$, $\hat{\beta}_1 = 20$, $\hat{\beta}_2 = 0.07$, $\hat{\beta}_3 = 35$, $\hat{\beta}_4 = 0.01$, $\hat{\beta}_5 = -10$.

   (a) Which answer is correct, and why?

   i. For a fixed value of IQ and GPA, high school graduates earn more on average than college graduates.

   ii. For a fixed value of IQ and GPA, college graduates earn more on average than high school graduates.

   iii. For a fixed value of IQ and GPA, high school graduates earn more on average than college graduates provided that the GPA is high enough. iv. For a fixed value of IQ and GPA, college graduates earn more on

   average than high school graduates provided that the GPA is high enough.

   Solution: The model is:

   $y = \beta_0 + \beta_1 \cdot \text{GPA} + \beta_2 \cdot \text{IQ} + \beta_3 \cdot \text{Level} + \beta_4 \cdot \text{GPA} \cdot \text{IQ} + \beta_5 \cdot \text{GPA} \cdot \text{Level}$

   Fixing IQ and GPA, changing Level from 0 to 1 will change the outcome by:

   $\Delta y = \beta_3 + \beta_5 \cdot \text{GPA}$

   $\Delta y > 0 \Rightarrow \beta_3 + \beta_5 \cdot \text{GPA} > 0 \Rightarrow \text{GPA} < \dfrac{-\beta_3}{\beta_5} = \dfrac{-35}{-10} = 3.5$

   Option iii correct.

   (b) Predict the salary of a college graduate with IQ of 110 and a GPA of 4.0.

   (c) True or false: Since the coefficient for the GPA/IQ interaction term is very small, there is very little evidence of an interaction effect. Justify your answer.

   Solution: This is false. It is important to remember that GPA and IQ vary over different scales. It is better to explicitly test the significance of the interaction effect, and/or visualize or quantify the effect on sales under realistic ranges of GPA/IQ values.

4. I collect a set of data ($n = 100$ observations) containing a single predictor and a quantitative response. I then fit a linear regression model to the data, as well as a separate cubic regression, i.e. $Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \beta_3 X^3 + \epsilon$.

   (a) Suppose that the true relationship between X and Y is linear, i.e. $Y = \beta_0 + \beta_1 X + \epsilon$. Consider the training residual sum of squares (RSS) for the linear regression, and also the training RSS for the cubic regression. Would we expect one to be lower than the other, would we expect them to be the same, or is there not enough information to tell? Justify your answer.

   Solution: You would expect the cubic regression to have lower RSS since it is at least as flexible as the linear regression.

   (b) Answer (a) using test rather than training RSS.

   Solution: Though we could not be certain, the test RSS would likely be higher due to overfitting.

(c) Suppose that the true relationship between X and Y is not linear, but we don't know how far it is from linear. Consider the training RSS for the linear regression, and also the training RSS for the cubic regression. Would we expect one to be lower than the other, would we expect them to be the same, or is there not enough information to tell? Justify your answer.

Solution: You would expect the cubic regression to have lower RSS since it is at least as flexible as the linear regression.

(d) Answer (c) using test rather than training RSS.

Solution: There is not enough information to tell, it depends on how non-linear the true relationship is.

5. Consider the fitted values that result from performing linear regression without an intercept. In this setting, the ith fitted value takes the form

$$\hat{y}_i = x_i \hat{\beta},$$

where

$$\hat{\beta} = \left(\sum_{i=1}^{n} x_i y_i \right) / \left(\sum_{i'=1}^{n} x_{i'}^2 \right).$$

show that we can write

$$\hat{y}_i = \sum_{i'=1}^{n} a_{i'} y_{i'}$$

What is $a_{i'}$?

Note: We interpret this result by saying that the fitted values from linear regression are linear combinations of the response values.

Solution:

$$\hat{y}_i = x_i \frac{\sum_{i=1}^{n} x_i y_i}{\sum_{i'=1}^{n} x_{i'}^2} \tag{1}$$

$$= x_i \frac{\sum_{i'=1}^{n} x_{i'} y_{i'}}{\sum_{i''=1}^{n} x_{i''}^2} \tag{2}$$

$$= \frac{\sum_{i'=1}^{n} x_i x_{i'} y_{i'}}{\sum_{i''=1}^{n} x_{i''}^2} \tag{3}$$

$$= \sum_{i'=1}^{n} \frac{x_i x_{i'} y_{i'}}{\sum_{i''=1}^{n} x_{i''}^2} \tag{4}$$

$$= \sum_{i'=1}^{n} \frac{x_i x_{i'}}{\sum_{i''=1}^{n} x_{i''}^2} y_{i'} \tag{5}$$

therefore,

$$a_{i'} = \frac{x_i x_{i'}}{\sum x^2}$$

6. Using (3.4), argue that in the case of simple linear regression, the least squares line always passes through the point $(\bar{x}, \bar{y})$.

Solution: When $x = \bar{x}$ what is y?

$$y = \hat{\beta}_0 + \hat{\beta}_1 \bar{x} \tag{6}$$
$$= \bar{y} - \hat{\beta}_1 \bar{x} + \hat{\beta}_1 \bar{x} \tag{7}$$
$$= \bar{y} \tag{8}$$

7. It is claimed in the text that in the case of simple linear regression of Y onto X, the R^2 statistic (3.17) is equal to the square of the correlation between X and Y (3.18). Prove that this is the case. For simplicity, you may assume that $\bar{x} = \bar{y} = 0$.

   Solution: We have the following equations:

$$R^2 = \frac{TSS - RSS}{TSS}$$
$$Cor(x, y) = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_i (x_i - \bar{x})^2} \sqrt{\sum_i (y_i - \bar{y})^2}}$$

   As above, its important to remember $\sum_i x_i = \sum_j x_j$
   when $\bar{x} = \bar{y} = 0$

$$Cor(x, y)^2 = \frac{(\sum_i x_i y_i)^2}{\sum_i x_i^2 \sum_i y_i^2}$$

   Also note that:

$$\hat{y}_i = \hat{\beta}_o + \hat{\beta}_1 x_i = x_i \frac{\sum_j x_j y_j}{\sum_j x_j^2}$$

   Therefore, given that $RSS = \sum_i (y_i - \hat{y}_i)^2$ and $TSS = \sum_i (y_i - \bar{y})^2 = \sum_i y_i^2$

$$R^2 = \frac{\sum_i y_i^2 - \sum_i (y_i - x_i \frac{\sum_j x_j y_j}{\sum_j x_j^2})^2}{\sum_i y_i^2} \tag{9}$$
$$= \frac{\sum_i y_i^2 - \sum_i (y_i^2 - 2 y_i x_i \frac{\sum_j x_j y_j}{\sum_j x_j^2} + x_i^2 (\frac{\sum_j x_j y_j}{\sum_j x_j^2})^2)}{\sum_i y_i^2} \tag{10}$$
$$= \frac{2 \sum_i (y_i x_i \frac{\sum_j x_j y_j}{\sum_j x_j^2}) - \sum_i (x_i^2 (\frac{\sum_j x_j y_j}{\sum_j x_j^2})^2)}{\sum_i y_i^2} \tag{11}$$
$$= \frac{2 \sum_i (y_i x_i) \frac{\sum_j x_j y_j}{\sum_j x_j^2} - \sum_i (x_i^2) \frac{(\sum_j x_j y_j)^2}{(\sum_j x_j^2)^2}}{\sum_i y_i^2} \tag{12}$$
$$= \frac{2 \frac{(\sum_i x_i y_i)^2}{\sum_j x_j^2} - \frac{(\sum_i x_i y_i)^2}{\sum_j x_j^2}}{\sum_i y_i^2} \tag{13}$$
$$= \frac{(\sum_i x_i y_i)^2}{\sum_i x_i^2 \sum_i y_i^2} \tag{14}$$

Chapter 4: Classification

1. Here we predict a qualitative (categorical) variable rather than a quantitative one. This is known as classification rather than regression.
2. There are many possible classification techniques we will discuss including:
   (a) Logistic regression
   (b) Linear discriminant analysis
   (c) Quadratic discriminant analysis
   (d) Naive Bayes
   (e) K-nearest neighbors
   (f) Start of generalized linear models (Poisson regression)
   (g) More methods in later chapters (generalized additive models, trees, random forests, boosting, and support vector machines

4.1: An overview of classification

1. Just as in regression, our classification models will have a set of training observations $(x_1, y_1), \ldots, (x_n, y_n)$. Only now y_i will be a categorical variable.
2. Example: Auto dataset high / low mpg classification by binning mpg quant variable. Note especially does not generalize to 3+ classes, though all pairing of classes could be considered.

4.2: Why not linear regression?

1. Even if we encode random variable Y numerically such as $0, 1, \ldots$ for each category, a regression model would predict values outside the range of Y. Further, ordering categories numerically makes little sense. Hard issues when residuals are not normal, and variance of residuals is not constant. Coefs are not meaningful or reliable. Performance also is bad.
2. Instead, we aim to give a meaningful estimate of conditional probability $Pr(Y|X)$.

4.3: Logistic regression

1. The logistic model:
   (a) For simplicity, consider the 2 class case encoded as 0 and 1.
   (b) How to model the relationship between $p(X) = Pr(Y = 1|X)$ and X?
   (c) We want a function $0 \leq p(X) \leq 1$ which makes sense as a probability. Many options.
   (d) The logistic function is most popular:

$$p(X) = \frac{e^{\beta_0+\beta_1 X}}{1 + e^{\beta_0+\beta_1 X}}$$

   It has an "s" shape and is symmetric around point $(0, 0.5)$. Also it is defined for any value of X. Extreme values of X give probabilities close to zero. This is a "soft" version of a step function with nice math properties.

(e) Origins of the logistic curve: For probability p, if you model odds (success to failure ratio) as

$$odds = \frac{p}{1-p}$$

which lies in $[0, \infty)$. If betting, would say 5:1 odds of rolling a 2. If win, would get 1+5 dollars, else lose 1 dollar. Then, the log of odds gives

$$\text{logit}(p) = \ln\left(\frac{p}{1-p}\right).$$

Now this transformation has range $(-\infty, \infty)$ making it suitable for linear regression.

$$\text{logit}(p) = \beta_0 + \beta_1 x \quad \Rightarrow \quad p = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x)}}$$

which is the logistic function.

(f) We will fit this logistic curve via maximum likelihood in the next section.

(g) Considering the rewritten log-odds form

$$\ln\left(\frac{p(X)}{1-p(X)}\right) = \beta_0 + \beta_1 X$$

we see a unit increase in X results in changing the log-odds by β_1 (increase/decrease based on sign). Equivalently, the odds changes by multiplier e^{β_1} (increase/decrease based on sign). Harder to see, the probability p changes according to the sigmoid function (increase/decrease based on sign).

2. Estimating the regression coefficients:

(a) Our estimates $\hat{\beta}_0, \hat{\beta}_1$ of β_0, β_1 are chosen such that they maximize the (Bernoulli) likelihood function

$$\ell(\beta | \{Y_i, X_i\}) = \ell(\beta_0, \beta_1) = \prod_{i:y_i=1} p(x_i) \prod_{i':y_{i'}=0} (1 - p(x_{i'}))$$

which assumes are training data is independent so joint probability because products. Note that

$$P(Y_i = 1 | X_i) = p(X_i), \quad P(Y_i = 0 | X_i) = 1 - p(X_i)$$

Mathematically, the log-likelihood function is rather used for numerical optimization.

$$\ln(\ell(\beta)) = \sum_{i=1}^{n} [y_i \ln(p_i) + (1 - y_i) \ln(1 - p_i)]$$

where p_i is the logistic function form.

(b) Intuitively, this amounts to choosing $\hat{\beta}_0, \hat{\beta}_1$ such that plugging these into the model $p(X)$ yields a number close to one for all positive occurrences and 0 otherwise.

(c) Maximum likelihood is a general approach which can be used for many non-linear models where least squares cannot be used. In fact, lease squares is a special case of maximum likelihood.

3. Making predictions: Estimates are easy once $\hat{\beta}$ are found. Same as linear regression only now results are interpreted as probabilities. A probability threshold (such as 0.5) is use to convert probability to category.

(a) **Model:** Logistic regression models the log-odds linearly:

$$\text{logit}\big(P(Y=1 \mid X_1,\ldots,X_p)\big) = \log\frac{P(Y=1 \mid X)}{P(Y=0 \mid X)} = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p$$

(b) **Decision boundary:** Defined by $P(Y=1 \mid X) = 0.5$:

$$0.5 = \frac{1}{1+e^{-(\beta_0+\beta_1 X_1+\cdots+\beta_p X_p)}} \quad \Rightarrow \quad \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p = 0$$

(c) **Interpretation:** This is a linear equation in the predictors, i.e., a hyperplane in p-dimensional space. The logistic function introduces nonlinearity in probabilities, but the 0.5 contour (decision boundary) remains linear.

(d) **Nonlinear boundaries:** To obtain nonlinear decision boundaries, one can transform predictors before modeling, e.g., include X^2, interactions X_1X_2, or use splines.

4. Multiple logistic regression:

(a) Extension is straightforward:

$$p(X) = \frac{e^{\beta_0+\beta_1 X_1+\cdots+\beta_p X_p}}{1+e^{\beta_0+\beta_1 X_1+\cdots+\beta_p X_p}} \quad \Rightarrow \quad \ln\left(\frac{p(X)}{1-p(X)}\right) = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p$$

(b) Again, maximum likelihood is used to estimate parameters.

5. Multinomial logistic regression: Extend to the $K > 2$ classifier. 2 key desires: all probabilities sum to 1, and model log odds with a linear function.

(a) Select the Kth class as the baseline case. Then our new model is

$$Pr(Y=k|X=x) = \frac{e^{\beta_{k0}+\beta_{k1}x_1+\ldots\beta_{kp}x_p}}{1+\sum_{l=1}^{K-1} e^{\beta_{l0}+\beta_{l1}x_1+\cdots+\beta_{lp}x_p}}$$

for $k = 1, 2, \ldots, k-1$, and

$$Pr(Y=K|X=x) = \frac{1}{1+\sum_{l=1}^{K} e^{\beta_{l0}+\beta_{l1}x_1+\cdots+\beta_{lp}x_p}}.$$

This last equation results from forcing all probabilities to sum to 1. Same with the summation in the denominator.

(b) One can show that

$$\ln\left(\frac{Pr(Y=k|X=x)}{Pr(Y=K|X=x)}\right) = \beta_{k0} + \beta_{k1}x_1 + \ldots\beta_{kp}x_p$$

for $k = 1, 2, \ldots, K-1$.

(c) This shows the log odds between any pair of classes is linear in the features. The odds in this case are any class compared to the Kth one.

(d) An alternative coding for multinomial logistic regression is known as the softmax coding. It is effectively equivalent to the above (same fitted values, log odds, key model outputs, etc). Instead we have no baseline class and one more β coefficient to estimate.

$$Pr(Y=k|X=x) = \frac{e^{\beta_{k0}+\beta_{k1}x_1+\cdots+\beta_{kp}x_p}}{\sum_{l=1}^{K} e^{\beta_{l0}+\beta_{l1}x_1+\cdots+\beta_{lp}x_p}}.$$

$$\ln\left(\frac{Pr(Y=k|X=x)}{Pr(Y=k'|X=x)}\right) = (\beta_{k0}-\beta_{k'0}) + (\beta_{k1}-\beta_{k'1})x_1 + \ldots(\beta_{p0}-\beta_{p'0})x_p$$

4.4: Generative models for classification

1. Logistic regression directly models the conditional probability $Pr(Y = k|X = x)$ via the sigmoid function. This is a modeling of a conditional distribution.
2. An alternate approach instead models the distribution of predictors X separately in each response class Y, then uses Baye's theorem to flip these around into estimates for $Pr(Y = k|X = x)$.
   (a) When distributions are normal, this aligns very closely with logistic regression.
   (b) This approach is called "generative" because they learn the data generating process for each class, not just a decision boundary.
   (c) Synthetic data for class can be generated from the learned distributions.
3. Why bother having another model when we already have logistic regression?
   (a) When classes are substantially separate, logistic regression is unstable. This approach solves this.
   (b) For small sample and normal distributions of X by classes Y, this approach can be more accurate.
4. Recall Bayes' Theorem:
   (a) Basic version:
   $$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$
   where
   $$\begin{aligned} P(A \mid B) &= \text{probability of event } A \text{ given that } B \text{ occurred (posterior)} \\ P(B \mid A) &= \text{probability of event } B \text{ given } A \text{ occurred (likelihood)} \\ P(A) &= \text{probability of } A \text{ occurring (prior)} \\ P(B) &= \text{total probability of } B \text{ occurring (normalizing constant), often computed as} \\ P(B) &= P(B \mid A)P(A) + P(B \mid A^c)P(A^c) \end{aligned}$$
   Example on probability of getting an apple given it is shiny. Bag of 0.7 apple, 0.3 orange. 0.8 apples shiny. 0.1 oranges shiny. Update probability based on new information. Move from one cond prob to another.
   (b) Classification version:
   $$\begin{aligned} P(Y = 1 \mid X) &= \frac{P(X \mid Y = 1)\, P(Y = 1)}{P(X \mid Y = 1)P(Y = 1) + P(X \mid Y = 0)P(Y = 0)}, \\ P(Y = 0 \mid X) &= \frac{P(X \mid Y = 0)\, P(Y = 0)}{P(X \mid Y = 1)P(Y = 1) + P(X \mid Y = 0)P(Y = 0)}. \end{aligned}$$
   We aim to model the density functions $P(X|Y = 1)$ and $P(X|Y = 0)$. Decision rule: classify X as 1 if $P(Y = 1 \mid X) > P(Y = 0 \mid X)$, otherwise 0.
5. Approach: Simple example
   (a) **Data:**

Class	0	1
x	$1, 2$	$4, 5$

(b) **Class means:**

$$\mu_0 = \frac{1+2}{2} = 1.5, \quad \mu_1 = \frac{4+5}{2} = 4.5$$

(c) **Shared variance:**

$$\sigma^2 = \frac{(1-1.5)^2 + (2-1.5)^2 + (4-4.5)^2 + (5-4.5)^2}{4} = 0.25$$

(d) **Class priors:**

$$P(y=0) = P(y=1) = 0.5$$

(e) **Class-conditional Gaussian likelihoods:**

$$P(x \mid y=k) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu_k)^2}{2\sigma^2} \right), \quad k = 0, 1$$

(f) **Posterior probability for $x = 3.2$:**

$$P(y=1 \mid x=3.2) = \frac{P(x=3.2 \mid y=1)\, P(y=1)}{P(x=3.2 \mid y=0)\, P(y=0) + P(x=3.2 \mid y=1)\, P(y=1)} \approx 0.916$$

(g) **Decision boundary:** Generally a linear function in X.

6. Approach: Suppose Y has K classes.

   (a) Let $\pi_k = Pr(Y = k)$ denote the prior probability that an observation comes from class k.

   (b) Let $f_k(X) = Pr(X|Y = k)$ denote the density function of X for an observation that comes from the kth class. That is, $f_k(x)$ is large if there is a high probability that an observation in the kth class has $X \approx x$, and small otherwise.

   (c) Then, Bayes' theorem states

$$p_k(x) = Pr(Y = k|X = x) = \frac{\pi_k f_k(x)}{\sum_{l=1}^{K} \pi_l f_l(x)}$$

   where $p_k(x)$ is the posterior probability that an observation $X = x$ belongs to the kth class. Note also that the denominator is expanded form of $Pr(X = x)$ by classes of Y as the total law of probability.

   (d) Estimating π_k is easy, just fraction of training observations in each class.

   (e) Estimating density function $f_k(x)$ is much more challenging, leading to discriminant analysis here. Once we can estimate $f_k(x)$, this approach because an approximation of the Bayes classifier described in Chapter 2.

7. Linear discriminant analysis: for $p = 1$, one predictor case

   (a) Assume $f_k(x)$ is a normal distribution (AKA Gaussian).

$$f_k(x) = \frac{1}{\sqrt{2\pi}\sigma_k} \exp\left(-\frac{1}{2\sigma_k^2}(x-\mu_k)^2 \right)$$

   where μ_k and σ_k are the mean and variance parameters for the kth class.

   (b) Assumed a shared variance across all classes $\sigma = \sigma_1 = \cdots = \sigma_K$.

(c) Pluggin into $p_k(x)$, we have

$$p_k(x) = Pr(Y = k|X = x) = \frac{\pi_k \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{1}{2\sigma_k^2}(x-\mu_k)^2\right)}{\sum_{l=1}^{K} \pi_l \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{1}{2\sigma^2}(x-\mu_l)^2\right)}$$

(d) Assign $X = x$ the class for which $p_k(x)$ is largest. Taking the natural log of $p_k(x)$, this equates to maximizing

$$\delta_k(x) = x \cdot \frac{\mu_k}{\sigma^2} - \frac{\mu_k^2}{2\sigma_2} + \log(\pi_k).$$

Note this is a linear function in x.

(e) For instance, if $K = 2$ and $\pi_1 = \pi_2$, then the Bayes classifier assigns observation to class 1 if $2x(\mu_1 - \mu_2) > \mu_1^2 - \mu_2^2$ and to class 2 otherwise. The Bayes decision boundary is the point for which $\delta_1 = \delta_2$. This becomes

$$x = \frac{\mu_1^2 - \mu_2^2}{2(\mu_1 - \mu_2)} = \frac{\mu_1 + \mu_2}{2}.$$

(f) Assuming normal class distributions, we still need an estimate for μ_k and σ_k. Linear discriminant analysis (LDA) does this from the training set.

$$\hat{\mu}_k = \frac{1}{n_k} \sum_{i:y_i=k} x_i$$

$$\hat{\sigma} = \frac{1}{n-K} \sum_{k=1}^{K} \sum_{i:y_i=k} (x_i - \hat{\mu}_k)^2$$

(g) In the end, LDA assigns the class k to $X = x$ for which

$$\hat{\delta}_k(x) = x \cdot \frac{\hat{\mu}_k}{\hat{\sigma}^2} - \frac{\hat{\mu}_k^2}{2\hat{sigma}^2} + \ln(\hat{\pi}_k)$$

is maximized. Not again the linear function of x.

8. Linear discriminant analysis for $p > 1$:

(a) Extend LDA to the case of multiple predictors, $X = (X_1, \ldots, X_p)$, by drawing from a multivariate Gaussian.

(b) Review of multivariate Gaussian:

- Assume each predictor follows a one-dimensional normal distribution with some correlation between each pair of predictors.
- If $p = 2$, we have a surface in 3D. The height of this surface at a point represents the probability that both X_1 and X_2 fall in a small region around that point.
- Notation: A p-dimensional random variable X has a multivariate Gaussian distribution is written $Z \sim N(\mu, \Sigma)$ where $E(X) = \mu$ is the mean of X (a vector of p components), and $Cov(X) = \Sigma$ is the $p \times p$ covariance matrix of X.
- Formally, the multivariate Gaussian density is

$$f(x) = \frac{1}{(2\pi)^{p/2}|\Sigma|^{1/2}} \exp\left(-\frac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu)\right)$$

- The kth classis drawn from its own multivariate Gaussian $N(\mu_k, \Sigma)$ where the covariance matrix is still common to each class.
- The density function becomes $f_k(X = x)$ and we assign $X = x$ a class for which

$$\delta_k(x) = x^T \Sigma^{-1} \mu_k - \frac{1}{2} \mu_k^T \Sigma^{-1} \mu_k + \ln(\pi_k)$$

is largest.

4.5: A comparison of classification models

1. An analytical comparison: LDA vs QDA vs naive Bayes vs logistic regression

   (a) Consider these approaches in a setting with K classes, so that we assign an observation to the class that maximizes $Pr(Y = k|X = x)$. We set K as the baseline class and assign an observation to the class that maximizes

$$\ln\left(\frac{Pr(Y = k|X = x)}{Pr(Y = K|X = x)}\right)$$

   for $k = 1, \ldots, K$. Examine this form for each method.

   (b) Method comparison:

   - Logistic regression

$$\ln\left(\frac{Pr(Y = k|X = x)}{Pr(Y = K|X = x)}\right) = \beta_{k0} + \sum_{j=1}^{p} \beta_{kj} x_j$$

   - LDA

$$\ln\left(\frac{Pr(Y = k|X = x)}{Pr(Y = K|X = x)}\right) = a_k + \sum_{j=1}^{p} b_{kj} x_j$$

     where a_k and b_k depend on π_k, μ_k, and Σ. Note this is a linear form just as in logistic regression

   - QDA

$$\ln\left(\frac{Pr(Y = k|X = x)}{Pr(Y = K|X = x)}\right) = a_k + \sum_{j=1}^{p} b_{kj} x_j + \sum_{j=1}^{p} \sum_{l=1}^{p} c_{kjl} x_j x_l$$

     similar to LDA but with the quadratic interaction terms

   - Naive Bayes

$$\ln\left(\frac{Pr(Y = k|X = x)}{Pr(Y = K|X = x)}\right) = \beta_{k0} + \sum_{j=1}^{p} \beta_{kj} x_j$$

4.6: Generalized linear models

Skip

4.bonus: Classification evaluation metrics

1. Metric definitions:

	Predicted Positive	**Predicted Negative**
Actual Positive	True Positive (TP)	False Negative (FN)
Actual Negative	False Positive (FP)	True Negative (TN)

Table 3: Confusion matrix for binary classification.

Metric	**Formula**	**Interpretation**
Accuracy	$\frac{TP+TN}{TP+TN+FP+FN}$	Overall fraction correct
Error Rate	$1 - \text{Accuracy}$	Overall fraction incorrect
Sensitivity (Recall, TPR)	$\frac{TP}{TP+FN}$	Ability to detect positives, completeness
Specificity (TNR)	$\frac{TN}{TN+FP}$	Ability to detect negatives
Precision	$\frac{TP}{TP+FP}$	Reliability of pos predictions, pos more important, exactness
F1 Score	$2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$	Balance of precision and recall, harmonic mean, is a percent
ROC Curve	Plot: Sensitivity vs. $1 - \text{Specificity}$	Tradeoff across thresholds, receiver operating char curve
AUC	Area under ROC curve	Overall discrimination ability

Table 4: Common evaluation metrics for classification models.

	Predicted Positive	**Predicted Negative**
Actual Positive	40 (TP)	10 (FN)
Actual Negative	20 (FP)	30 (TN)

Table 5: Confusion matrix example with 100 total observations.

2. Example: Spam email detection

Computed Metrics:

$$\begin{aligned}
\text{Accuracy} &= \frac{40+30}{100} = 0.70 \\
\text{Error Rate} &= 1 - 0.70 = 0.30 \\
\text{Sensitivity (Recall, TPR)} &= \frac{40}{40+10} = 0.80 \\
\text{Specificity (TNR)} &= \frac{30}{30+20} = 0.60 \\
\text{Precision} &= \frac{40}{40+20} = 0.67 \\
\text{F1 Score} &= 2 \cdot \frac{0.67 \cdot 0.80}{0.67+0.80} \approx 0.73
\end{aligned}$$

3. ROC and AUC:

- **ROC Curve (Receiver Operating Characteristic):** Plots **True Positive Rate (TPR)** vs. **False Positive Rate (FPR)** as the classification threshold varies.

$$\text{TPR (Sensitivity, Recall)} = \frac{TP}{TP+FN}, \quad \text{FPR} = \frac{FP}{FP+TN} = 1 - \text{Specificity}$$

- **Interpretation:**
  - Each point represents a specific threshold for classifying positives.

  - Curve starts at (0,0) (predict everything negative) and ends at (1,1) (predict everything positive).
  - Ideal classifier: curve hugs the top-left corner.
  - Random classifier: diagonal line from (0,0) to (1,1).
- **AUC (Area Under Curve):** Single-number summary of ROC curve performance.

$$0.5 \leq \text{AUC} \leq 1.0$$

- AUC = 1: perfect separation of classes - AUC = 0.5: random guessing - Higher AUC = better model

- **Advantages:**
  - Threshold-independent evaluation.
  - Works well even with class imbalance.
  - Visualizes trade-off between sensitivity and specificity.

Lab: Classification methods

Homework

1. Using a little bit of algebra, prove that (4.2) is equivalent to (4.3). In other words, the logistic function representation and logit representation for the logistic regression model are equivalent.

   Solution: We need to show that

$$p(X) = \frac{e^{\beta_0+\beta_1 X}}{1 + e^{\beta_0+\beta_1 X}}$$

   is equivalent to

$$\frac{p(X)}{1 - p(X)} = e^{\beta_0+\beta_1 X}$$

   Letting $x = e^{\beta_0+\beta_1 X}$

$$\frac{P(X)}{1 - p(X)} = \frac{\frac{x}{1+x}}{1 - \frac{x}{1+x}} \tag{15}$$

$$= \frac{\frac{x}{1+x}}{\frac{1}{1+x}} \tag{16}$$

$$= x \tag{17}$$

2. When the number of features p is large, there tends to be a deterioration in the performance of KNN and other local approaches that perform prediction using only observations that are near the test observation for which a prediction must be made. This phenomenon is known as the curse of dimensionality, and it ties into the fact that non-parametric approaches often perform poorly when p is large. We will now investigate this curse.

   (a) Suppose that we have a set of observations, each with measurements on $p = 1$ feature, X. We assume that X is uniformly (evenly) distributed on $[0, 1]$. Associated with each observation is a response value. Suppose that we wish to predict a test observation's

response using only observations that are within 10% of the range of X closest to that test observation. For instance, in order to predict the response for a test observation with $X = 0.6$, we will use observations in the range $[0.55, 0.65]$. On average, what fraction of the available observations will we use to make the prediction?

Solution: For values in 0...0.05, we use less than 10% of observations (between 5% and 10%, 7.5% on average), similarly with values in 0.95...1. For values in 0.05...0.95 we use 10% of available observations. The (weighted) average is then $7.5 \times 0.1 + 10 \times 0.9 = 9.75\%$.

(b) Now suppose that we have a set of observations, each with measurements on $p = 2$ features, X_1 and X_2. We assume that (X_1, X_2) are uniformly distributed on $[0, 1] \times [0, 1]$. We wish to predict a test observation's response using only observations that are within 10range of X_1 and within 10% of the range of X_2 closest to that test observation. For instance, in order to predict the response for a test observation with $X_1 = 0.6$ and $X_2 = 0.35$, we will use observations in the range $[0.55, 0.65]$ for X_1 and in the range $[0.3, 0.4]$ for X_2. On average, what fraction of the available observations will we use to make the prediction?

Solution: Since we need the observation to be within range for X_1 and X_2 we square 9.75% $= 0.0975^2 \times 100 = 0.95\%$

(c) Now suppose that we have a set of observations on $p = 100$ features. Again the observations are uniformly distributed on each feature, and again each feature ranges in value from 0 to 1. We wish to predict a test observation's response using observations within the 10% of each feature's range that is closest to that test observation. What fraction of the available observations will we use to make the prediction?

Solution: Similar to above, we use: $0.0975^{100} \times 100 = 8 \times 10^{-100}\%$, essentially zero.

(d) Using your answers to parts (a)–(c), argue that a drawback of KNN when p is large is that there are very few training observations "near" any given test observation.

Solution: As p increases, the fraction of observations near any given point rapidly approaches zero. For instance, even if you use 50% of the nearest observations for each p, with $p = 10$, only $0.5^{10} \times 100 \approx 0.1\%$ points are "near".

(e) Now suppose that we wish to make a prediction for a test observation by creating a p-dimensional hypercube centered around the test observation that contains, on average, $10p = 1, 2$, and 100, what is the length of each side of the hypercube? Comment on your answer.

Note: A hypercube is a generalization of a cube to an arbitrary number of dimensions. When $p = 1$, a hypercube is simply a line segment, when $p = 2$ it is a square, and when $p = 100$ it is a 100-dimensional cube.

Solution: When $p = 1$, clearly the length is 0.1. When $p = 2$, we need the value l such that $l^2 = 0.1$, so $l = \sqrt{0.1} = 0.32$. When $p = n$, $l = 0.1^{1/n}$, so in the case of $n = 100$, $l = 0.98$. Therefore, the length of each side of the hypercube rapidly approaches 1 (or 100%) of the range of each p.

3. Suppose we collect data for a group of students in a statistics class with variables $X_1 =$ hours studied, $X_2 =$ undergrad GPA, and $Y =$ receive an A. We fit a logistic regression and produce estimated coefficient, $\hat{\beta}_0 = -6$, $\hat{\beta}_1 = 0.05$, $\hat{\beta}_2 = 1$.

(a) Estimate the probability that a student who studies for 40h and has an undergrad GPA of 3.5 gets an A in the class.

Solution: :The logistic model is:

$$\log\left(\frac{p(X)}{1 - p(x)}\right) = -6 + 0.05X_1 + X_2$$

or

$$p(X) = \frac{e^{-6+0.05X_1+X_2}}{1+e^{-6+0.05X_1+X_2}}$$

when $X_1 = 40$ and $X_2 = 3.5$, $p(X) = 0.38$

(b) How many hours would the student in part (a) need to study to have a 50% chance of getting an A in the class?

Solution: We would like to solve for X_1 where $p(X) = 0.5$. Taking the first equation above, we need to solve $0 = -6 + 0.05X_1 + 3.5$, so $X_1 = 50$ hours.

4. Suppose that we take a data set, divide it into equally-sized training and test sets, and then try out two different classification procedures. First we use logistic regression and get an error rate of 20% on the training data and 30% on the test data. Next we use 1-nearest neighbors (i.e. $K = 1$) and get an average error rate (averaged over both test and training data sets) of 18%. Based on these results, which method should we prefer to use for classification of new observations? Why?

   Solution: For $K = 1$, performance on the training set is perfect and the error rate is zero, implying a test error rate of 36%. Logistic regression outperforms 1-nearest neighbor on the test set and therefore should be preferred.

5. This problem has to do with odds.

   (a) On average, what fraction of people with an odds of 0.37 of defaulting on their credit card payment will in fact default?

   Solution: Odds is defined as $p/(1-p)$.

$$0.37 = \frac{p(x)}{1-p(x)}$$

therefore,

$$p(x) = \frac{0.37}{1+0.37} = 0.27$$

   (b) Suppose that an individual has a 16% chance of defaulting on her credit card payment. What are the odds that she will default?

   Solution:

$$0.16/(1-0.16) = 0.19$$

6. Suppose that you wish to classify an observation $X \in \mathbb{R}$ into 'apples' and 'oranges'. You fit a logistic regression model and find that

$$\hat{Pr}(Y = orange|X = x) = \frac{\exp(\hat{\beta}_0 + \hat{\beta}_1 x)}{1+\exp(\hat{\beta}_0 + \hat{\beta}_1 x)}$$

Your friend fits a logistic regression model to the same data using the softmax formulation in (4.13), and finds that

$$\hat{Pr}(Y = orange|X = x) = \frac{\exp(\hat{\alpha}_{orange0} + \hat{\alpha}_{orange1} x)}{\exp(\hat{\alpha}_{orange0} + \hat{\alpha}_{orange1} x) + \exp(\hat{\alpha}_{apple0} + \hat{\alpha}_{apple1} x)}$$

(a) What is the log odds of 'orange' versus 'apple' in your model?
Solution: The log odds is just $\hat{\beta}_0 + \hat{\beta}_1 x$

(b) What is the log odds of 'orange' versus 'apple' in your friend's model?
Solution: From 4.14, log odds of our friend's model is:

$$(\hat{\alpha}_{orange0} - \hat{\alpha}_{apple0}) + (\hat{\alpha}_{orange1} - \hat{\alpha}_{apple1})x$$

(c) Suppose that in your model, $\hat{\beta}_0 = 2$ and $\hat{\beta} = -1$. What are the coefficient estimates in your friend's model? Be as specific as possible.
Solution: We can say that in our friend's model $\hat{\alpha}_{orange0} - \hat{\alpha}_{apple0} = 2$ and $\hat{\alpha}_{orange1} - \hat{\alpha}_{apple1} = -1$.
We are unable to know the specific value of each parameter however.

(d) Now suppose that you and your friend fit the same two models on a different data set. This time, your friend gets the coefficient estimates $\hat{\alpha}_{orange0} = 1.2$, $\hat{\alpha}_{orange1} = -2$, $\hat{\alpha}_{apple0} = 3$, $\hat{\alpha}_{apple1} = 0.6$. What are the coefficient estimates in your model?
Solution: The coefficients in our model would be $\hat{\beta}_0 = 1.2 - 3 = -1.8$ and $\hat{\beta}_1 = -2 - 0.6 = -2.6$

(e) Finally, suppose you apply both models from (d) to a data set with 2,000 test observations. What fraction of the time do you expect the predicted class labels from your model to agree with those from your friend's model? Explain your answer.
Solution: The models are identical with different parameterization so they should perfectly agree

Chapter 5: Resampling methods

1. 2 common resampling methods:
   (a) Cross-validation: estimate test error to evaluate performance (model assessment), or select model flexibility (model selection)
   (b) Bootstrapping: Measure accuracy of a parameter estimate or model

5.1: Cross-validation

1. Usually, test data sets are not available. We only have access to a single training set. Unfortunately, training error often dramatically underestimates test error. So, this must be remedied. Recall u-shape test error as flexibility increases from chapter 2.
2. Here we estimate the test error by making use of a hold-out of the training data, though we still train the final model with all the data we have. Approaches discussed will apply to both regression and classification problems.
3. The validation set approach:
   (a) Randomly divide the training set into 2 parts: Training set and validation (hold-out) set. We fit the model on the training set and assess accuracy on the validation set as a way to approximate test error. In the end though, we still use the full data for training our final model.
   (b) Repeating this random validation set process many times gives many approximations of test error.

(c) Drawbacks:

- Validation estimates of test error can vary a lot.
- Because we reduce the training set size, our model will perform worse than it would if we trained on the full data. For that reason, validation error will overestimate the true training error.

4. Leave-one-out cross-validation (LOOCV):

(a) Reserve a single observation for the validation set. Repeat for all possible validation sets.

(b) The test MSE is then estimated by the average of all model errors.

$$CV_{(n)} = \frac{1}{n}\sum_{i=1}^{n} MSE_i, \quad MSE_i = (y_i - \hat{y}_i)^2$$

Note that each MSE is from a different model here.

(c) Advantages:

- LOOCV has less bias and tends not to overestimate the true test error.
- Individual MSE will vary less because each model uses most of the observations as training data.

(d) A major disadvantage is LOOCV is computationally expensive. In the case of least squares linear or polynomial regression, there is a nice result.

$$CV_{(n)} = \frac{1}{n}\sum_{i=1}^{n} \frac{(y_i - \hat{y}_i)^2}{1 - h_i}$$

where $\hat{y}_i$ is the prediction of the full model trained on all data, and h_i is the leverage term from page 99. This is almost the ordinary MSE with correction $1/n < h_i < 1$, which reflects the amount an observation influences its own fit.

5. k-fold cross validation:

(a) Randomly divide the set of observations into k groups (or folds). Use the first fold as validation, the rest as training. MSE_1 is the resulting test accuracy. Repeat for all k folds. Then,

$$CV_{(k)} = \frac{1}{k}\sum_{i=1}^{k} MSE_i.$$

(b) LOOCV is a special case of k-fold cross validation, though the most computational.

(c) There is a bias-variance tradeoff in the choice of k.

(d) Goal of the flexibility MSE curves is either to (1) find the minimum to identify model flexibility, or (2) estimate test error.

6. Bias-variance trade-off for k-fold cross validation:

(a) Aside from efficiency, k fold cross validation has another advantage over LOOCV: bias-variance trade-off. With bias reduction, LOOCV has the win because models are trained with most of that data. But, for variance, k-fold CV has the lower result because its models are less correlate to to eachother.

7. Cross-validation on classification problems:

(a) CV extends naturally here as you would expect. LOOCV error rate takes the form

$$CV_{(n)} = \frac{1}{n}\sum_{i=1}^{n} Err_i$$

where $Err_i = I(y_i \neq \hat{y}_i)$.

5.2: The bootstrap

1. For models like linear regression, one can derive standard error (SE) formulas for confidence interval and hypothesis tests. Alternatively, bootstrapping can be used to estimate SE. This can be applied to complex models where SE formulas aren't available. Statkey demo?

Lab: Cross-validation and the bootstrap

Homework

Chapter 6: Linear model selection and regularization

1. Recall (again, and again): The bias-variance tradeoff:

   $$\text{Expected Test Error} = \text{Bias}^2 + \text{Variance} + \text{Var}(\varepsilon) = \big(\mathbb{E}[\hat{f}(x)] - f(x)\big)^2 + \mathbb{E}\big[(\hat{f}(x) - \mathbb{E}[\hat{f}(x)])^2\big] + \text{Var}(\varepsilon)$$

   (a) Balance between a model's simplicity and its flexibility.

   (b) Bias measures how much a model's predictions systematically differ from the true values.
   - High bias → model is too simple (e.g., underfitting).
   - It makes strong assumptions and misses important patterns in the data.

   (c) Variance measures how much a model's predictions change when trained on different samples of the data.
   - High variance → model is too complex (e.g., overfitting).
   - It fits noise in the training data, leading to poor generalization.

   (d) Tradeoff: As model complexity increases...
   - Bias decreases (the model fits training data better).
   - Variance increases (the model becomes more sensitive to noise).
   - The goal is to find a sweet spot—a model complex enough to capture the true pattern (low bias) but simple enough to generalize well (low variance).

2. The standard linear model for regression is

   $$Y = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p + \epsilon$$

   which is typically fitted via least squares. We will keep this linear framework for now because of its many advantages (simplicity, interpretability, etc), but we will improve by considering alternate fitting techniques.

3. Advantages of alternative fitting:

   (a) Better predictive accuracy: When $n >> p$, least squares estimates have low variance and hence perform better on test observations. If n is not much bigger than p, there can be a lot of variability, resulting in overfitting, and poor accuracy. Our strategy will be to constrain (or shrink) the estimated coefficients.

(b) Better model interpretability: When there are many variables, some may be irrelevant leading to unnecessary complexity. We remove such variables.

4. Summary of options:

   (a) Subset selection: Identify a best subset of p predictors and remove rest. Choose the best model via cross validation.

   (b) Shrinkage: Keep all predictors, but shrink the coefficients towards zero to remove variance.

   (c) Dimension reduction: Project p variables onto a M-dimensional subspace and use these instead (skip for now).

6.1: Subset selection

1. Best subset selection:

   (a) Fit separate least squares regressions for each possible combinations of the p predictors. Note, there are 2^p possibilities.

   (b) Algorithm: $p + 1$ possible model options.

   - Let M_0 denote the null model with no predictors.
   - For $k = 1, 2, \ldots, p$: Fit all $\binom{p}{k}$ models with k predictors. Pick the best, call it M_k using smallest RSS (or other option).
   - Select a single best model from $M_0, \ldots, M_p$ using prediction error on a validation set (or other options).

2. Stepwise selection:

   (a) Best subset selection is not feasible computationally for large p. There are also statistical issues. The larger the search space, the higher the chance of finding models that look good on the training data even though they will suffer on the test data (overfitting and high variance of coefficient estimates).

   (b) Forward stepwise selection:

   - Start with no predictors, add best predictors (most model improvement) one at a time until all predictors are in the model.
   - Reduction to $1 + \sum_{k=1}^{p-1}(p-k) = 1 + p(p+1)/2$ models.
   - Choose from all such models via cross validation.

   (c) Backward stepwise selection:

   - Start with all predictors, remove the least best predictor (least model improvement) one at a time until all predictors are in the model.
   - Reduction to $1 + \sum_{k=1}^{p-1}(p-k) = 1 + p(p+1)/2$ models.
   - Choose from all such models via cross validation.

   (d) Hybrid: Combine the 2.

   -

3. Choosing the optimal model:

   (a) How to tell which model is best? Previously, we said the best model has the smallest RSS and the largest R^2, since these qualities are related to training error. Instead, we wish to choose a model with low test error. Because training error can be a poor estimate of test error, we need a new approach.

(b) 2 common approaches:

- Indirectly estimate test error by making and adjustment to the training error to account for the bias due to overfitting.
- Directly estimate the test error using either a validation set or cross validation approach.

(c) C_p, AIC, BIC, and adjusted R^2: Adjusting the training error for the model size.

- C_p: Estimate test MSE (recall, $MSE = RSS/n$) as

$$C_p = \frac{1}{n}(RSS + 2d\hat{sigma}^2).$$

where $\hat{sigma}^2$ is an estimate of the variance of the error ϵ associated with each response measurement, typically using the full model containing all predictors.

- Essentially, the C_p statistic adds a penalty of $2d\hat{\sigma}^2$ to the training RSS to adjust for the fact that the training error tends to underestimate the test error. This penalty increases as the number of predictors in the model increases balancing a corresponding decrease in RSS. Once can show C_p is an unbiased estimate of the test MSE.
- AIC (Akaike information criterion):

(d) AIC (Akaike information criterion):

- AIC is defined for a large class of models fit by maximum likelihood.
- In our case, it is given by

$$AIC = \frac{1}{n}(RSS + 2d\hat{sigma}^2)$$

where for simplicity we have omitted irrelevant constants.

- Once can see AIC and C_p are proportional to eachother.
- Choose lowest AIC.

(e) BIC (Bayes information criterion):

- BIC is derived from a Bayesian point of view, but ends up looking similar to C_p and AIC.
- For the least squares model with d predictors, BIC is given up to irrelevant constants as

$$BIC = \frac{1}{n}(RSS + \ln(n)d\hat{\sigma}^2$$

- BIC places a higher penalty for models with many variables compared to C_p, resulting in the selection of smaller models.
- Choose lowest BIC.

(f) Adjusted R^2:

- Good for comparing models with different numbers of variables.
- Recall,

$$R^2 = 1 - \frac{RSS}{TSS}, \quad TSS = \sum(y_i - \overline{y})^2, \quad \text{the total sum of squares}$$

The R^2 always increases as more variables are added.

- For a least squares model with d variables,

$$\text{Adjusted } R^2 = 1 - \frac{RSS/(n-d-1)}{TSS/(n-1)}$$

- Large value of adjusted R^2 indicates a model with small test error.
- Intution is that once all of the correct variables have been included in the model, adding additional noise variables leads to only a very small decrease in RSS and the increase in d will drag down the overall value.

(g) Validation and cross-validation:

- As an alternative to these metrics, we can directly estimate the test error via validation and CV.
- Advantage is we directly estimate the test error and make fewer assumptions about the underlying model.
- The only drawback is computational efficiency.
- Can select a model using the on-standard-error rule: calculate the stard error of the estimated test MSE for each model size, then select the smallest model for which the estimated test error is within one standard error of the lowest point on the curve allowing for choice of the simplest model.

6.2: Shrinkage methods

With linear regression, we minimize below residual sum of squares.

$$RSS = \sum_i (y_i - \hat{y}_i)^2 = \sum_i \left(y_i - \beta_0 - \sum_j \beta_j x_{ij} \right)^2$$

In the case of overfitting (eg n close to p), β_j coefficients have high variance which translates into large values. Shrinkage methods shrink these unruly coefficients towards zero with the main aim of preventing overfitting. Two related strategies: ridge regression and the lasso. This approach is also known generally as regularization.

1. Ridge regression)(ridge shape shows on the cost function):

   (a) Estimate regression coefficients β_i^R (note not the same as OLS) by minimizing a modified RSS formula:

   $$RSS + \lambda \sum_j \beta_j^2 = \sum_i \left(y_i - \beta_0 - \sum_j \beta_j x_{ij} \right)^2 + \lambda \sum_j \beta_j^2$$

   where $\lambda \geq 0$ is a tuning parameter usually found via cross validation. Note λ term is only for $j \geq 1$, not to the intercept term β_0.

   (b) Ridge regression tradeoff:

   i. Minimize RSS just as with OLS.
   ii. $\lambda \sum \beta_j$ is a shrinkage penalty which forces $\beta_1, \ldots, \beta_n$ close to zero. Larger λ makes shrinkage more severe and reducing RSS less important.
   iii. This in reality is a set of regression coefficients, β_λ^R, one per parameter λ. Finding the right λ is key.
   iv. Extreme cases: $\lambda = 0$ gives the same as OLS. $\lambda \to \infty$ gives null model (no predictors, just intercept).

   (c) Example figures from text: Test error and bias/variance tradeoff. Note ℓ^2 norm $\| \cdot \|_2$ is just Euclidean distance. Note MSE of OLS model is almost the same as the null model.

(d) Warning: OLS gives the same β_j coefficients regardless of X_j scale. Not so for ridge regression and β_j^R's can change substantially. Normalizing data is key practice. Z-score works or equation (6.6) in text.

(e) In the case with $n < p$, less samples than predictors, OLS cannot be fit. Though, ridge regression still applies.

(f) In general, ridge regression is best when least squares estimates has high variance.

(g) Finally, ridge regression is much more efficient than any form of variable selection.

2. The lasso (Least Absolute Shrinkage and Selection Operator):

(a) Ridge regression has one disadvantage. It keeps all coefficients / variables present which makes it harder to interpret. This new approach addresses this.

(b) Estimate regression coefficients β_i^L (note not the same as OLS) by minimizing a modified RSS formula:

$$RSS + \lambda \sum_j |\beta_j| = \sum_i \left(y_i - \beta_0 - \sum_j \beta_j x_{ij} \right)^2 + \lambda \sum_j |\beta_j|$$

where $\lambda \geq 0$ is a tuning parameter usually found via cross validation. Note λ term is only for $j \geq 1$, not to the intercept term β_0.

(c) This new λ term involves the ℓ_1 norm: $\|\beta\| = \sum_j |\beta_j|$. This is the taxicab norm which can be seen in 2d.

(d) This penalty norm allows for β_λ^L terms to be zero, meaning lass does both shrinkage and variable selection at the same time resulting in simpler models.

3. Who wins? Ridge regression or LASSO?

(a) No clear winner, let CV decide.

(b) If only a small subset of predictors are actually predicting response variable y, then LASSO has the advantage.

(c) May be ridge by change.

(d) May be similar, in which case, choose the simpler of model.

(e) In general, ridge is more stable (less coef variance) and smooth (small changes in MSE) while LASSO is simpler. There is a hybrid approach called elastic net.

$$RSS + \lambda_1 \sum_j |\beta_j| + \lambda_2 \sum_j \beta_j^2 = \sum_i \left(y_i - \beta_0 - \sum_j \beta_j x_{ij} \right)^2 + \lambda_1 \sum_j |\beta_j| + \lambda_2 \sum_j \beta_j^2$$

(f) Note, alt version as constrained optimization. Shows in the homework.

$$\text{minimize} \sum_i \left(y_i - \beta_0 - \sum_j \beta_j x_{ij} \right)^2, \quad \text{subject to} \sum_j \beta_j^2 \leq s$$

$$\text{minimize} \sum_i \left(y_i - \beta_0 - \sum_j \beta_j x_{ij} \right)^2, \quad \text{subject to} \sum_j |\beta_j| \leq s$$

6.3: Dimension reduction methods

6.4: Considerations in higher dimensions

Lab: Linear models and regularization methods

Homework

Exam 1 summary: Chapters 1-6

1. Chapter 1: Introduction
   (a) General intro, no exam content
2. Chapter 2: Statistical learning
   (a) Goal: Relate response Y to predictors $X_1, \ldots, X_p$. Regression case:
   $$Y = f(X) + \epsilon$$
   (b) 2 key tasks: Prediction ($Y \approx \hat{Y} = \hat{f}(X)$) and inference (understanding X and Y relationship).
   (c) Tradeoffs: Flexibility vs interpretability, bias-variance tradeoff
   (d) Error: MSE for train and test dataset
   (e) Key formulas:
      i. Reducible and irreducible error, holding $f(X), \hat{f}(X)$ constant: $E[(Y - \hat{Y})^2] = (f(X) - \hat{f}(X))^2 + Var(\epsilon)$
      ii. Bias-variance tradeoff: $E[(Y_0 - \hat{f}(x_0))^2] = [Bias(\hat{f}(x_0))]^2 + Var(\hat{f}(x_0)) + Var(\epsilon)$ where $Bias(\hat{f}(x_0)) = E[\hat{f}(x_0)] - f(x_0)$.
      iii. "U" shaped test data fit curve. Over vs underfitting.
   (f) Parametric vs nonparametric models
   (g) Regression vs classification
   (h) Supervised vs unsupervised learning
   (i) Classification error, Bayes classifier.
3. Chapter 3: Linear regression
   (a) Simple linear regression
      i. Coef values from minimizing residual sum of squares
      ii. Coef confidence intervals and hypothesis tests involving SE formulas
      iii. Fit via residual standard error (RSE) and R^2.
   (b) Multiple linear regression
      i. Coef values from minimizing residual sum of squares, not nice formula
      ii. Coef confidence intervals and hypothesis tests involving SE formulas. F-statistic.
      iii. Fit via residual standard error (RSE) and R^2.
   (c) 4 diagnostic plots
4. Chapter 4: Classification
   (a) Predict a qualitative output, key idea is to estimate class probabilities

(b) Logistic regression, log-odds as a linear function.

$$\ln(p/(1-p)) = \beta_0 + \beta_1 X_1 + \dots$$

(c) Evaluation via confusion matrix, accuracy, precision, recall, F1 score

(d) ROC curve, AUC for many thresholds

(e) Multiple logistic regression extension

(f) LDA and QDA

5. Chapter 5: Resampling methods

   (a) Cross validation

   (b) Bootstrap

6. Chapter 6: Linear model selection and regularization

   (a) Variable selection

   (b) Shrinkage methods (regularization)

Chapter 7: Moving beyond linearity

Skip

7.1: Polynomial regression

7.2: Step functions

7.3: Basis functions

7.4: Regression splines

7.5: Smoothing splines

7.6: Local regression

7.7: Generalized additive models

Lab: Non-linear modeling

Homework

Chapter 8: Tree-based methods

Tree based methods apply to both regression and classification problems. They involve stratifying or segmenting the predictor space into a number of simple regions. A single tree is simple and interpreable, though they are inferior to other learning methods when it comes to accuracy. For that reason, this chapter also introduces ensembles (consensus prediction of many trees) including bagging, random forest, boosting, and others.

8.1: Basics of decision trees

Regression trees: Predict a quantitative variable Y via predictors $X_1, X_2, \ldots, X_P$

1. Key idea:
   (a) Divide the predictor data X into regions $R_1, R_2, \ldots, R_J$ such that within each region, the response Y can be predicted by the mean of the observations in that region.

   $$\hat{y} = \text{average of } y_i \text{ for } X \text{ that lies within that region}$$

   (b) Model generated:

   $$\hat{f}(X) = \sum_{j=1}^{J} \hat{y}_{R_j} I(x \in R_j)$$

2. Tree building: Recursive binary splitting
   (a) Start with data all in one region.
   (b) For each predictor X_j and each possible split point s:
   - Divide the data into 2 regions:

     $$R_1(j,s) = \{X | X_j < s\}, \quad R_2(j,s) = \{X | X_j \geq s\}$$

     where cutpoints s are taken as midpoints of training data predictor variables.
   - Compute the residual sum of squares:

     $$RSS(j,s) = \sum_{i: x_i \in R_1} (y_i - \hat{y}_{R_1})^2 + \sum_{i: x_i \in R_2} (y_i - \hat{y}_{R_2})^2$$

   - Choose the predictor X_j and cutpoint s that minimizes RSS.
   - Repeat within each resulting region until some stopping criteria is met (eg min number of training samples in a node).
   - Top of tree is the root, bottom are leaves, middle points are nodes, rest are branches.
3. Predictions: Once the tree is built, for a new observation x_0
   (a) Trace x_0 down the tree until lands at a leaf.
   (b) The predicted response $\hat{y}$ is the mean of all training responses y_i in that leaf.
4. Overfitting and pruning
   (a) A fully grown tree (only one training sample per node) will likely overfit.
   (b) Trees are pruned to balance over / under fitting.
   (c) Cost-complexity pruning acts like regularization with α chosen via cross-validation.

   $$R_\alpha(T) = \sum_{m=1}^{|T|} \sum_{i \in R_m} (y_i - \hat{y}_{R_m})^2 + \alpha |T|$$

   where $|T|$ is the number of terminal nodes (leaves). Growing a tree fully then performing CV over α values is doable, while looking at all possible 2^d subtrees.
5. Interpretation

(a) Tree structure easy visual

(b) Top nodes most influential

(c) Terminal regions give groups of observations with similar responses

6. Adv / disadvs:

   (a) Advantages: Simple, interpretable, handles nonlinearity and interactions automatically, handle categorical variables

   (b) Disadvantages: Unstable (small changes to train data can have large effects), low accuracy in general.

7. Classification trees: Predict a qualitative Y via predictors $X_1, X_2, \ldots, X_p$.

   (a) Approach: Almost the same as the regression case above.

   - Predict $\hat{y}(x_0)$ as the majority class of region R_j containint x_0.
   - Compare partition candidates for predictor X_j and cutpoint s ($R_1(j,s) = \{X|X_j < s\}$ and $R_2(j,s) = \{X|X_j \geq s\}$) via impurity measure of each region as a weighted average (see below).
   - Apply cost-complexity pruning via cross validation.

$$R_\alpha(T) = \sum_{m=1}^{|T|} Q_m(T) + \alpha|T|$$

   where $Q_m(T)$ is the impurity measure at node m, and α penalizes tree size.

   (b) Impurity measures: Choose the best split by weighted measures. Denote $\hat{p}_{mk}$ as the proportion of training observations in node m belonging to class k. The purity of node m is either:

   - Classification error: $1 - \max_k \hat{p}_{mk}$
   - Gini index: Low is more pure

$$G_m = \sum_k \hat{p}_{mk}(1 - \hat{p}_{mk}) = 1 - \sum_k \hat{p}_{mk}^2$$

     This measures the total variance across the K classes. Small if $\hat{p}_{mk}$ is close to zero or 1. Measures how often you'd misclassify a random observation in that node if you randomly labeled it according to the node's class proportions. 50/50 and 90/10 examples.

   - Cross-entropy (deviance):

$$-\sum_k \hat{p}_{mk} \log(\hat{p}_{mk})$$

     Comes from information theory and penalizes uncertainty more heavily than Gini. Note because $\hat{p}_{mk} \geq 0$, then $-\hat{p}_{mk}\log(\hat{p}_{mk}) \geq 0$. Like Gini, entropy is near zero if all $\hat{p}_{mk}$ are near 0 or 1.

8. Tree vs linear regression: Parametric vs non parametric, bias vs flexible, interpretable

   (a) Linear regression: $Y = f(x) + \epsilon = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p + \epsilon$

   (b) Regression tree: $Y = f(x) + \epsilon = \sum c_m I(X \in R_m) + \epsilon$ where c_m is the region mean.

8.2: Bagging, random forests, boosting, and Bayesian additive regression trees

An ensemble method combines many simple models (aka weak learners) in order obtain a single, possibly very powerful model. Draw picture of idea with notes of regression and classification cases.

1. Bagging (general approach across statistical learning)
   (a) Standard decision trees tend to suffer from high variance (small changes in train data can have large model impact), though they have low bias. Bootstrap aggregation (or bagging) is a strategy to reduce this variance while keeping bias low. For example, n independent observations $Z_1, \ldots, Z_n$ each with variance σ, the variance of the mean $\overline{Z}$ has new variance σ^2/n (averaging observations reduces variance).
   (b) Bagging ensemble of trees:
   - Grow many trees on bootstrap samples. For training size n, draw B bootstrap samples. Denote each fitted tree as $\hat{f}^{*b}(x)$ and the final prediction as

$$\hat{f}_{bag}(x) = \frac{1}{B}\sum_b \hat{f}^{*b}(x), \quad \text{or} \quad \text{majority vote}\left\{\hat{f}^{*b}(x)\right\}$$

   - Average the tree results (for regression) or use majority vote (for classification) for prediction. Simulate test error on out-of-bag (OOB) samples left out of bootstrapping (usually around 1/3) removing the need for CV.
   - Variance is reduced compared to a single tree.
   - Importance of predictors is measured by how much each variable reduces impurity on average. Large values are better.

2. Random forest
   (a) Modification of bagging that decorrolates trees. A top predictor will always be at the root of a tree giving very similar models in bagging.
   (b) Each split only considers a random subset of predictors $m \approx \sqrt{p}$.
   (c) Result is usually better performing than bagging. See text figure.

3. Boosting (general approach across statistical learning)
   (a) Build trees sequentially, where each new tree focuses on residuals of the old one. Keep correcting mistakes.
   (b) Boosting for regression (squared-error loss):
   - Start with $\hat{f}(x) = 0$ (null model)
   - For $b = 1, 2, \ldots, B$, fit a small regression tree $\hat{f}^b(x)$ to the train data (X, r) (note residuals in place of Y), and update the model

$$\hat{f}(x) = \hat{f}(x) + \lambda \hat{f}^b(x)$$

     and update the residuals

$$r_i = r_i - \lambda \hat{f}^b(x_i)$$

   - Final model:

$$\hat{f}(x) = \sum_{b=1}^{B} \lambda \hat{f}^b(x)$$

   (c) Tuning parameters:

- Number of trees B (boosting iteration)
- Shrinkage λ (learning rate)
- Tree depth d (splits per tree)

(d) For classification: Minimize log-loss instead of squared error with the aim to improve distinguishing between classes.

(e) Bias-variance tradeoff: Reduces both bias and variance.

- Bias: Very low, many trees can capture complex patterns)
- Variance: Controlled by small trees and shrinkage
- Overall: Often outperforms bagging and random forest in test error, if tuned properly.

Lab: Decision trees

Homework

Chapter 9: Support vector machines

Here we leverage hyperplanes for classification. Results is a classic CS ML technique from 1990s which often performs well. We build the ideas by growing from simple to complex:

- Draw picture of each
- Maximal margin classifier (optimal hyperplane separation if possible)
- Support vector classifier (allow misclassification at a cost)
- Support vector machine (nonlinear separation via kernel trick)

9.1: Maximal margin classifier

1. Goal is to find a hyperplane ($p-1$ dimension affine subspace of $\mathbb{R}^p$) that best separates 2 classes in a dataset.

   (a) Hyperplane in 2D: $\beta_0 + \beta_1 X_1 + \beta_2 X_2 = 0$ (line in plane).

   (b) Hyperplane in 3D: $\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 = 0$ (plane in 3 space).

   (c) Hyperplane in pD: $\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \cdots + \beta_p X_p = 0$.

   (d) Short notation: $f(X) = \beta_0 + \vec{\beta}^T \vec{X} = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \cdots + \beta_p X_p$

   (e) A hyperplane divides the space into 2. Evaluating $f(X)$ at a point not on the hyperplane gives either -1 or 1 depending on which side it is on. This will be used for classification.

   (f) Recall, geometry of a hyperplane. For line l in 2D, draw an example. Define in terms of vectors as all $\vec{x}$ such that $\vec{w} \cdot (\vec{x} - \vec{x}_0) = 0$ (only need a point an vector orthogonal to hyperplane). Expanding this gives the same form as above where $\vec{w} = \vec{\beta}$ and $\vec{w} \cdot \vec{x}_0 = \beta_0$.

2. A hyperplane separates p-dimensional space into 2 regions. If the data is perfectly separable, there are many options. Draw 2D case. Not always possible of course.

   (a) For training data $x_i = (x_{i1}, x_{i1}, \ldots, x_{ip})^T$ with classes $y_i \in \{-1, 1\}$ for $i = 1, 2, \ldots, n$, a hyperplane separates classes y_i if

$$\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p > 0, \quad \text{if } y_i = 1$$

and

$$\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p < 0, \quad \text{if } y_i = -1$$

which can be combined as

$$y_i \left(\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p\right) > 0, \quad \text{for all } i$$

written more compactly as

$$y_i \left(\beta_0 + \vec{\beta}^T \vec{x_i}\right) > 0, \quad \text{for all } i$$

(b) For a new test datapoint x^*, we can classify based on the sign of $f(x^*) = \beta_0 + \beta_1 x_1^* + \cdots + \beta_p x_P^* = \beta_0 + \vec{\beta}^T \vec{x^*}$ where the magnitude of $f(x^*)$ can give a measure of certainty.

3. Maximal margin classifier separates the classes with the largest possible margin (distance between hyperplane and closest data points). Points closest to the boundary are the support vectors.

(a) The margin of the hyperplane is the smallest perpendicular distance from the training data to the hyperplane.

$$M = \min_i \frac{y_i(\beta_0 + \vec{\beta}^T \vec{x}_i)}{\|\vec{\beta}\|}$$

where the perpendicular distance from a point x_i to the hyperplane is

$$\frac{|\beta_0 + \vec{\beta}^T \vec{x}_i|}{\|\vec{\beta}\|}.$$

(b) Find the maximal margin via the optimization problem:

$$\max_{\beta_i, M} M, \quad \text{subject to } \sum_j \beta_j^2 = 1$$

$$y_i \left(\beta_0 + \beta_1 x_{i1} + \ldots \beta_p x_{ip}\right) \geq M, \quad \text{all } i$$

Note, the full margin is $2M$, one M on either side of the hyperplane.

(c) Maximizing the margin makes the model robust to small pertubations in training data. Only the support vectors effect the model, so this classifier has low variance in general.

(d) If separation is not possible, this optimization model fails. Also note how small changes in the train data relative to the support vectors can have large impact on the hyperplane and margin.

9.2: Support vector classifiers

1. If perfect separation is not possible, we add a slack variable $\epsilon \geq 0$ to allow for misclassifications. In this way, the support vector classifier is also known as a soft margin classifier.

2. Optimization problem:

$$\max_{\beta_i, M} M, \quad \text{subject to } \sum_j \beta_j^2 = 1$$

$$y_i \left(\beta_0 + \beta_1 x_{i1} + \ldots \beta_p x_{ip}\right) \geq M(1 - \epsilon_i), \quad \text{all } i$$

$$\epsilon_i \geq 0, \quad \sum \epsilon_i \leq C$$

where C is a nonnegative tuning parameter (large C allows for margin and hyperplane violations) which gives bias variance tradeoffs (small C has high bias, low variance and large C visa versa).

3. ϵ_i has utility:
   (a) $\epsilon_i = 0$ operates as before.
   (b) $\epsilon_i > 0$ means the point is on the wrong side of the margin.
   (c) $\epsilon_i > 1$ means the point is on the wrong side of the hyperplane.
   (d) At most C of the ϵ_i can be on the wrong side of the hyperplane.
4. Only data on or violating the margin will effect the classifier decision boundary. These are the support vectors.
5. Drawback of everything above: Decision boundaries are linear due to the design of the hyperplane separation. Pictures in text.

9.3: Support vector machines

1. To generalize beyond linear separation, we need to grow the feature space.
   (a) Include polynomial features: $X_i, X_i^2, X_iX_j, X_i^3$, etc
   (b) Quadratic decision boundary example in 2D: $f(X) = \beta_0 + \beta_1X_1 + \beta_2X^2 + \beta_3X_1^2 + \beta_4X_2^2 + \beta_5X_1X_2 = 0$ (quadratic conic sections)
   (c) Show general optimization in the text. Similar to support vector classifier. Issue is high degree polynomials get complex, so we resolve this with a Kernel function approach.
2. The kernel trick:
   (a) Recall: Inner product or dot product.

$$\langle x, y\rangle = \sum_{j=1}^{p} x_j y_j = \|x\|\|y\| \cos(\theta)$$

   where θ is the angle between the vectors. This is a measure of similarity.
   (b) The support vector classifier can be written in terms of the training observations:

$$f(\mathbf{x}) = \beta_0 + \boldsymbol{\beta}^T\mathbf{x},$$

   where the coefficient vector can be expressed as

$$\boldsymbol{\beta} = \sum_{\mathbf{x}_j \in S} \alpha_j y_j \mathbf{x}_j.$$

   Substituting this expression for $\boldsymbol{\beta}$ gives

$$\begin{aligned} f(\mathbf{x}) &= \beta_0 + \left(\sum_{\mathbf{x}_j \in S} \alpha_j y_j \mathbf{x}_j\right)^T \mathbf{x} \\ &= \beta_0 + \sum_{\mathbf{x}_j \in S} \alpha_j y_j \langle \mathbf{x}, \mathbf{x}_j\rangle. \end{aligned}$$

   Here, S is the set of support vectors, $y_j \in \{-1, 1\}$ is the class label, and $\alpha_j > 0$ is the corresponding coefficient. All training observations that are not support vectors have $\alpha_j = 0$ and therefore do not contribute to the classifier.

(c) Replace the above inner product with a generalization of the form

$$K(x, x_i)$$

where K is a function known as a kernel and acts like a similarity measure.

(d) Common kernels:

- Above linear kernel: $K(x, x_i) = \langle x, x_i \rangle$ (hyperplane decision boundary)
- Degree d polynomial kernel: $K(x, x_i) = (1 + \langle x, x_i \rangle)^d$. Try for $p = d = 2$. (smooth curve)
- Radial kernel: $K(x, x_i) = \exp\left(-\gamma \sum_{j=1}^{p} (x_j - x_{ij})^2\right)$ for tuning parameter γ. (highly flexible curve, creates a bump like a Gaussian around each training point, large γ only influences neighbors/high variance while small goes farther/high bias)
- 2D examples in text.

(e) Example:

$$\begin{aligned} K(\mathbf{x}, \mathbf{x}_i) &= (1 + \mathbf{x}^T \mathbf{x}_i)^2 \\ &= (1 + x_1 x_{i1} + x_2 x_{i2})^2 \\ &= 1 + 2x_1 x_{i1} + 2x_2 x_{i2} \\ &\quad + x_1^2 x_{i1}^2 + 2x_1 x_2 x_{i1} x_{i2} + x_2^2 x_{i2}^2. \end{aligned}$$

This is equivalent to an inner product using the quadratic features x_1^2, $\sqrt{2} x_1 x_2$, and x_2^2, without explicitly constructing them.

9.4: SVMs with more than two classes

2 main approaches for $K > 2$ classes: Sample size matters here.

1. OVA: One vs all, train K 2-class models (one for each class, that class vs rest, classify to largest f value)
2. OVO: One vs one, train all $\binom{K}{2}$ paired models (classify to class which wins most paired comparisons). Best if K is not too big.

9.5: Relationship to logistic regression

Lab: Support Vector Machines

Homework

Chapter 10: Deep learning

Overview of ideas using text slides. Next deep dive into neural networks.
Links:

1. `https://upload.wikimedia.org/wikipedia/commons/f/f2/AI-History-Timeline-300dpi.jpg`
2. `https://www.tensorflow.org/`
3. `https://pytorch.org/`
4. `https://playground.tensorflow.org/`

5. https://projector.tensorflow.org/
6. https://deeplizard.com/resource/pavq7noze2
7. https://deeplizard.com/resource/pavq7noze3
8. https://mco-mnist-draw-rwpxka3zaa-ue.a.run.app/
9. https://poloclub.github.io/cnn-explainer/
10. https://magenta.tensorflow.org/sketch-rnn-demo
11. https://arxiv.org/abs/1607.06520
12. https://scikit-learn.org/stable/
13. https://www.deeplearning.ai/
14. https://experiments.withgoogle.com/experiments

10.1: Single layer neural networks

1. Use p predictors $X = (X_1, X_2, \ldots, X_p)$ to build nonlinear function $f(X)$ to predict response variable Y.
2. Single layer neural network structure: Draw an example with $p = 4, k = 5$
   (a) Input layer: X with p predictors.
   (b) Hidden layer: K activations (neurons, some neurons fire with data) A_k of the form

$$A_k = h_k(X) = g(w_{k0} + \sum_{j=1}^{p} w_{kj} X_j)$$

   where g is a nonlinear activation function such as sigmoid or ReLU. A_k has the role of (nonlinear) feature transformation.
   (c) Output layer: Linear combination of activations.

$$f(X) = \beta_0 + \sum_{k=1}^{K} \beta_k A_k$$

   (d) Parameters to estimate: β_k, w_{kj} which are fit by minimizing residual sum of squares of the train data

$$\sum_{i=1}^{n} (y_i - f(x_i))^2.$$

   in the regression case.
   (e) The weights w_{kj} are stored in a matrix W with $4 \times 5 = 20$ total values. The output layer has 6 more β_k coefs.
   (f) For classification, a softmax activation is used in the final layer, and negative multinomial log-likelihood (cross-entropy) is minimized.

10.2: Multilayer neural networks

Pictures and intuition in slides.

10.3: Convolution neural networks

Pictures and intuition in slides.

10.4: Document classification

Pictures and intuition in slides.

10.5: Recurrent neural networks

Pictures and intuition in slides.

10.6: When to use deep learning

No free lunch!

10.7: Fitting a neural network

1. The parameters to fit from the one layer example above are $\beta = (\beta_0, \beta_1, \ldots, \beta_K)$ and activation weights $w_k = (w_{k0}, w_{k1}, \ldots, w_{kp})$ for $k = 1, 2 \ldots, K$.
2. For our test data (x_i, y_i), we solve a nonlinear least squares problem:

$$\min_{w_k, \beta} \frac{1}{2} \sum_{i=1}^{n} (y_i - f(x_i))^2$$

   where

$$f(x_i) = \beta_0 + \sum_{k=1}^{K} \beta_k g \left(w_{k0} + \sum_{j=1}^{p} w_{kj} x_{ij} \right).$$

3. Writing all parameters in one long vector θ, we can rewrite the objective as

$$R(\theta) = \frac{1}{2} \sum_{i=1}^{n} (y_i - f_\theta(x_i))^2$$

   .

4. Minimizing $R(\theta)$ seem straightforward, but the complexity of the network paired with the nonlinear activation functions makes it more challenging. Such functions usually have local and global mins (draw pic). To resolve this and prevent overfitting, 2 strategies are used:
   (a) Slow learning: model is fit in a slow, iterative fashion via gradient descent which halts when overfitting occurs.
   (b) Regularization: Similar to lasso or ridge regression.
5. Gradient descent:
   (a) Start with initial guess θ^0, often random.
   (b) Iterate until the objective function $R(\theta)$ fails to decrease:
      - $\theta^{t+1} = \theta^t - \rho \nabla R(\theta^t)$ resulting in reduction in objective ($R(\theta^{t+1}) < R(\theta^t)$).
      - Small enuf ρ ensures decreasing R.
   (c) Determinant notation $\nabla R = \frac{\partial R}{\partial \theta}$ indicates the direction of steepest increase. Reversing this direction gives fasted decrease.

(d) Gradient calculation only requires the chain rule: $R(\theta) = \sum_{i=1}^{n} R_i(\theta) = \frac{1}{2}\sum_{i=1}^{n}(y_i - f_\theta(x_i))^2$.

$$R_i(\theta) = \frac{1}{2}\left(y_i - \beta_0 - \sum_{k=1}^{K}\beta_k g(w_{k0} + \sum_{j=1}^{p} w_{kj}x_{ij})\right)^2$$

Denote everything inside g as z_{ik}, then

$$\frac{\partial R_i(\theta)}{\partial \beta_k} = \frac{\partial R_i(\theta)}{\partial f_\theta(x_i)}\frac{\partial f_\theta(x_i)}{\partial \beta_k} = -(y_i - f_\theta(x_i))g(z_{ik})$$

and also

$$\frac{\partial R_i(\theta)}{\partial w_{kj}} = \frac{\partial R_i(\theta)}{\partial f_\theta(x_i)}\frac{\partial f_\theta(x_i)}{\partial g(z_{ik})}\frac{\partial g(z_{ik})}{\partial z_{ik}}\frac{\partial z_{ik}}{\partial w_{kj}} = -(y_i - f_\theta(x_i))\beta_k g'(z_{ik})x_{ij}$$

Note that a fraction of the residuals $(y_i - f_\theta(x_i))$ are assigned to the parameters via the chain rule. This is the heart of backward propagation.

6. To speed up gradient descent, a small fraction of the train sample (called a minibatch) is used to update parameters. This is known as stoichastic gradient descent.

7. Regularization akin to LASSO or ridge regression is used to prevent overfitting.

8. An epoch counts the number of times the equivalent of the training

10.8: Interpolation and double descent

Lab: Deep learning

Homework

Chapter 11: Survival analysis and censored data

Chapter 12: Unsupervised learning

Chapter 13: Multiple testing